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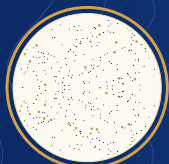
भारत का नवोद्यम केंद्र
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JADAVPUR UNIVERSITY

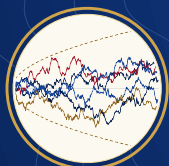
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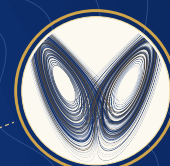
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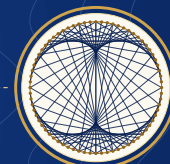
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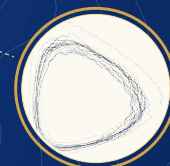
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ICMAAM 2026

ABSTRACT BOOK

4th INTERNATIONAL CONFERENCE ON

Mathematical Analysis and Applications in Modelling

12th – 14th OCTOBER 2026

Organized by

Department of Mathematics & CMBE, Jadavpur University

$$x_{n+1} = r x_n (1 - x_n) \quad \cdot \quad dX_t = f(X_t, X_{t-\tau}) dt + g(X_t) dW_t \quad \cdot \quad y^2 = x^3 + ax + b$$

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About the Conference

The **4th International Conference on Mathematical Analysis and Applications in Modelling (ICMAAM 2026)** is envisioned as an international meeting point for researchers working in mathematics, mathematical modelling, and closely connected scientific disciplines. The conference is intended to facilitate the presentation of recent research, encourage discussion on emerging mathematical techniques, and create opportunities for interaction among established academicians, early-career researchers, students, and professionals from different parts of the world.

The ICMAAM series began with **ICMAAM 2018 at Jadavpur University, Kolkata**, followed by its second edition, **ICMAAM 2023**, again hosted at Jadavpur University after the interruption caused by the COVID-19 pandemic. Building on the success of these editions, the conference expanded internationally with **ICMAAM 2024 at Chittagong University of Engineering and Technology (CUET), Bangladesh**, organized in collaboration with **Jadavpur University**, thereby further strengthening the academic reach and international character of the ICMAAM series. With this background, **ICMAAM 2026** brings the conference back to **Jadavpur University** for its fourth edition. The progression from the inaugural conference in 2018, through the second edition in 2023 and the international edition at CUET in 2024, reflects a sustained effort to develop ICMAAM into a forum where mathematical ideas can be discussed across institutional, disciplinary, and national boundaries. The 2026 conference seeks to carry this academic continuity forward while opening new avenues for research interaction and collaboration.

The scientific scope of ICMAAM 2026 covers a wide range of areas in **pure and applied mathematics, mathematical analysis, and mathematical modelling**. Topics of interest include, but are not restricted to, mathematical biology, epidemiological modelling, dynamical systems, ordinary and partial differential equations, stochastic processes and stochastic modelling, real and functional analysis, differential geometry, algebraic and topological methods, optimization, operations research, optimal control, parameter estimation, numerical analysis, scientific computing, computational fluid dynamics, biomechanics, theoretical and mathematical physics, complex systems, data-driven modelling, heat transfer, thermo-elasticity, and mathematical methods arising in engineering and the natural sciences. A major objective of the conference is to highlight the growing role of mathematics in the study of complex real-world systems. Contemporary scientific problems increasingly require the combined use of analytical theory, computation, modelling, data, and optimization. Mathematical frameworks now play a central role in understanding biological processes, disease transmission and treatment, ecological

dynamics, energy systems, industrial processes, environmental change, and many other phenomena characterized by interacting mechanisms and uncertainty.

Accordingly, ICMAAM 2026 encourages contributions that connect mathematical theory with practical scientific questions. Research involving deterministic and stochastic systems, control strategies, inverse problems, parameter estimation, numerical simulation, uncertainty analysis, and data-assisted modelling is expected to form an important part of the conference. Such work demonstrates how mathematical reasoning can contribute not only to the interpretation of complex phenomena but also to prediction, decision-making, intervention design, and technological development. The conference also seeks to provide an interdisciplinary setting in which mathematicians can interact with researchers from biology, medicine, physics, engineering, computation, environmental science, and related fields. Many of the most challenging scientific questions of the present era do not belong to a single discipline. Progress often depends on the ability to combine different perspectives, and mathematical modelling provides a natural framework through which such interdisciplinary connections can be developed.

Another important dimension of ICMAAM is the promotion of sustained international academic interaction. Scientific research advances most effectively when ideas, methods, and experiences are shared openly across institutions and countries. By bringing together participants with different academic backgrounds and research traditions, the conference aims to create opportunities for new collaborations, joint investigations, and long-term scholarly relationships. Particular attention is also given to the involvement of **young researchers and students**. Exposure to contemporary research and direct interaction with experienced scientists can significantly influence the development of early-career scholars. ICMAAM therefore seeks to create an environment in which younger participants can communicate their findings, receive constructive feedback, learn about new methodologies, and engage with researchers working at the forefront of their respective fields.

Beyond its academic programme, the conference recognizes that international scientific meetings also provide valuable opportunities for **cultural exchange and personal interaction**. Participants arriving from different regions and countries bring with them distinct academic traditions and cultural experiences. ICMAAM 2026 aims to create an atmosphere in which these differences enrich the conference experience and contribute to mutual understanding among the delegates. The cultural component associated with the conference is intended to introduce visiting participants to selected aspects of Indian and Bengali heritage through performances, music, literature, visual arts, and other forms of cultural expression. Such activities complement the scientific programme by allowing participants to experience the cultural environment of the host institution and the **city of Kolkata** while developing informal connections beyond the lecture halls.

The academic programme of **ICMAAM 2026** will include **Keynote Addresses (1), Plenary Lectures (13), Invited Talks (14), Young Scientist Symposium (11), Contributory Talks (78), and Poster Presentations (4)**. These sessions are designed to accommodate a broad range of research themes and to provide researchers at different stages of their careers with opportunities to present their work and participate in scientific discussion.

It is a privilege to host the fourth edition of ICMAAM at **Jadavpur University, Kolkata, India**. Jadavpur University has a long-standing tradition of scholarship and research across mathematics, science, engineering, humanities, and interdisciplinary studies. The **Department of Mathematics** and the **Centre for Mathematical Biology and Ecology** of **Jadavpur University** have contributed significantly to research in mathematical sciences and mathematical modelling and have been closely associated with the development of the ICMAAM conference series since its inception. The return of the conference to Jadavpur University in 2026 therefore carries a special significance. It reconnects the current edition with the origins of ICMAAM while also reflecting how the conference has grown through successive editions and international collaboration. The journey from **ICMAAM 2018 at Jadavpur University**, to **ICMAAM 2023 at Jadavpur University**, followed by **ICMAAM 2024 at CUET, Bangladesh**, and now **ICMAAM 2026**, illustrates the steady expansion of a scientific community built around common interests in mathematics, modelling, and interdisciplinary research.

As we look ahead to **ICMAAM 2026**, we envision a vibrant gathering of ideas, discoveries, and collaborations that will extend far beyond the days of the conference itself. We warmly welcome all participants to **Jadavpur University** and hope that the discussions initiated, connections established, and inspirations shared during this event will open new directions in mathematical research and foster lasting scientific partnerships across disciplines and borders. May ICMAAM 2026 serve not merely as a meeting of researchers, but as a meaningful step towards shaping the future of mathematics and its applications.

Prof. (Dr.) Priti Kumar Roy

Organizing Secretary

ICMAAM 2026

Day 1: 12th October, 2026

09.00 AM – 09.45 AM	Registration at TEQIP Building (1st floor), Jadavpur University	
9:45 AM : National Song		
9:50 AM : Inaugural Song		
9:55 AM : Lightening the Holy Lamp		
Venue: TEQIP Building (Room No: 208), Jadavpur University		
Formal Inaugural Session		
Chaired By	Prof. Chiranjib Bhattacharjee Vice Chancellor, Jadavpur University	
Welcome Address	Prof. Subhas Chandra Mandal Head, Department of Mathematics, Jadavpur University.	
About the Conference	Prof. Priti Kumar Roy Organizing Secretary, ICMAAM 2026	
Address by	Prof. Amitava Dutta Pro-Vice Chancellor, Jadavpur University	
Address by	International Guest	
Address by	Chief Guest	
Vote of Thanks	Prof. Dipak Kumar Kesh Joint Organizing Secretary, ICMAAM 2026	
Tea Break 10.45 am - 11.00 am		
Technical Session : Keynote Lecture (11:00am-11:45am) Chairperson: Prof. Joydeb Chattyopadhay TEQIP Building (Room No: 208)		
Name: Prof. František Štěpánek Department of Chemical Engineering, UCT Prague, Czech Republic	Title of Talk: Applications of mathematical modelling in pharmaceutical product and process development	Abstract Serial No. K-01
Plenary Talk 1 (11:45am-12:20pm) Chairperson: Prof. Kallol Paul TEQIP Building (Room No: 208)		
Name: Prof. Krishnendu Gongopadhyay IISER Mohali, India	Title of Talk: Twisted Patterson–Sullivan Densities on Nilpotent Covers for Anosov Subgroups.	Abstract Serial No. P-08

Plenary Talk 2 (12:20pm-12:55pm) Chairperson: Prof. Stacey Smith? TEQIP Building (Room No: 208)		
Name: Prof. Pralay Chatterjee IMSc, Chennai, India	Title of Talk: On the Betti numbers of a homogeneous space of a Lie group	Abstract Serial No. P-03
Plenary Talk 3 (12:55pm-01:30pm) Chairperson: Prof. Amit Chakraborty TEQIP Building (Room No: 208)		
Name: Prof. Igor Schreiber University of Chemical Technology, Prague	Title of Talk: Modelling of metabolic oscillations in cyanobacteria	Abstract Serial No. P-06
<u>Lunch Break (University Guest House)</u> 01:30pm – 02:30pm		
Invited Talks (02:30pm to 04:30pm) Parallel Sessions-I (Room No: 502)		
Invited Talk 1 (02:30pm to 03:00pm) Chairperson: Prof. Buddhadeb Sau Technology Bhavan (Room No: 502)		
Name: Prof. Fugo Takasu Nara Women's University, Japan	Title of Talk: Point pattern approach to study spatial population dynamics	Abstract Serial No. I-11
Invited Talk 2 (03:00pm to 03:30pm) Chairperson: Prof. Shovonlal Roy Technology Bhavan (Room No: 502)		
Name: Prof. Amit Chakraborty	Title of the talk: TopoSSR: A Topological Data Analysis Framework for Identifying Disease-Associated Protein Structural Mutations	Abstract Serial No. I-01
Invited Talk 3 (03:30pm to 04:00pm) Chairperson: Prof. Gopal Chandra Shit Technology Bhavan (Room No: 502)		

Name: Prof. Sarat Singh Mizoram University, India	Title of Talk: Lamb wave in the plate of microstretch thermoelastic diffusion materials	Abstract Serial No. I-03
Invited Talk 4 (04:00pm to 04:30pm) Chairperson: Prof. Prabir Kumar Kundu Technology Bhavan (Room No: 502)		
Name: Prof. Dharmendra Tripathi NIT Uttarakhand, India	Title of Talk: Mathematical Modelling of Tumor-Obstructed Non-Newtonian Biofluid Transport	Abstract Serial No. I-05
Invited Talks (02:30pm to 04:30pm) Parallel Sessions-II (Room No: 503)		
Invited Talk 5 (02:30pm to 03:00pm) Chairperson: Prof. Alaka Das Technology Bhavan (Room No: 503)		
Name: Prof. M. Sundarajan Mizoram University, India	Title of Talk: Quantification of Runoff and Sediment Transport for Sustainable Reservoir Management Using a Computational Modelling Approach	Abstract Serial No. I-02
Invited Talk 6 (03:00pm to 03:30pm) Chairperson: Prof. Tapan Kumar Kar Technology Bhavan (Room No: 503)		
Name: Dr. Bapan Ghosh IIT Indore, India	Title of Talk: Asymmetric Steady States and Population Recovery in a Delayed Patchy Model	Abstract Serial No. I-04
Invited Talk 7 (03:30pm to 04:00pm) Chairperson: Prof. Robert Hakl Technology Bhavan (Room No: 503)		
Name: Prof. Ujjwal Kumar Deb CUET, Bangladesh	Title of Talk: Effects of Solar Related Temperature Variations on Microalgae Cell Growth in a Photobioreactor	Abstract Serial No. I-10
Invited Talk 8 (4:00pm to 04:30pm) Chairperson: Prof. Amit Kumar Ghosh Technology Bhavan (Room No: 503)		
Name: Biplab Sinha Mahapatra	Title of the talk: Decision Analysis of Insurance Providers Using Double-Fold MEREC Weighting and MARCOS under Neutrosophic Information	Abstract Serial No. I-09

Young Scientist Symposium (2:30am-4:30am)		
Parallel Session-III		
Technology Bhavan (Room No: 402)		
Chairperson: Prof. Pritha Das		Time: 20 minutes (each)
Name: Dr. Shipan Chandra Deb CUET, Bangladesh	Title of Talk: Clinical Burden of Dual Pathogens: A comparative analysis of Mono infection Versus Zoonotic co-infection outcomes using SIR method	Abstract Serial No. S-04
Name: Dr. Ravi Kumar K IIT Ropar, India	Title of Talk: Rate-induced phase tipping in birhythmic oscillators	Abstract Serial No S-07
Name: Dr. Rajib Karmakar CUET, Bangladesh	Title of Talk: FEM utilizing the HHT-α Method for Nonlinear Dynamic Buckling of Imperfection-Sensitive Columns	Abstract Serial No S-11

Chairperson: Prof. Debashis Mukherjee		Time: 20 minutes (each)
Name: Dr. Amit Kumar Roy Jadavpur University, India	Title of Talk: Pulse-Induced Nonautonomous Dynamical Systems in Epidemiology: Modelling, Thresholds, and Disease Dynamics	Abstract Serial No. S-03
Name: Dr. Salil Ghosh Jadavpur University, India	Title of Talk: From Disease Dynamics to Sustainability: Mathematical Modelling of Complex Biological, Computational and Environmental Systems	Abstract Serial No S-09
Name: Dr. Subhankar Kushary Brainware University, India	Title of Talk: Role of Stem Cells in Skin Recovery: A Mathematical Approach to Understanding Psoriasis and Its Control	Abstract Serial No S-05

Tea Break (10 min)

Contributory Talk

Contributory Talk (04:40pm to 06:20pm)			
Parallel Sessions (C I/C II/C III/C IV)			
Parallel Session – C I– Analysis-Algebra (04:40pm to 05:30pm) Venue: Technology Bhavan - Room No.: 502 -5th floor Chairperson: Dr. Riddhick Birbonshi Time: 10 minutes (each)			
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Saikat Das	On Graded m-Nil Clean Rings	C-06
2.	Munna Pramanik	A study of planarity in bipartite graphs defined by group elements and subgroups	C-14
3.	Anjali Beniwal	On Factorization of Polynomial matrices	C-38
4.	Pranab Modak	Generating Sequence-Based SLRA for Toeplitz and Hankel Matrices	C-40
5.	Saikat Mahapatra	q-Berezin sectorial operators and their applications to q-Berezin number inequalities	C-10
Parallel Session – CI – Analysis-Combinatorics-Graph Theory (05:30pm to 06:20pm) Venue: Technology Bhavan - Room No.: 502 -5th floor Chairperson: Dr. Manashi Mandal Time: 10 minutes (each)			
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Gargi Kuila	On the Thickness of Line Graphs of Intersection Graphs of Ideals	C-03
7.	Parna Saha	On the Distance to Non-Primeness for Affine Structured Polynomial Matrices: Optimization Approach	C-41
8.	Anirban Tarafdar	A Graph-Based Mathematical Framework for Modelling Neurodegenerative Disease States from Patient-Level Data	C-77
9.	Sujan Khna	On Cauchy Split Continuous Functions	C-08

10.	Jayanta Manna	On an Improvement of the Blanco-Koldobsky-Turnšek Characterization of Isometries via Level Vectors	C-74
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Contributory Talk (04:40pm to 06:20pm) Parallel Sessions (C I/C II/C III/C IV)			
Parallel Session – C II- Mathematical Biology- Mathematical Modeling (04:40pm to 05:30pm)			
Venue: Technology Bhavan - Room No.: 503 -5th floor			
Chairperson: Dr. Santosh Biswas		Time: 10 minutes (each)	
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Purnendu Sardar	Patient-Data-Driven HIV Dynamics with Macrophage-Mediated Reinfection Using Noise-Induced Physics-Informed Neural Networks	C-02
2.	Imran Ali	Threshold Dynamics and Complex Behavior in a Nonlinear Impulsive Model of Oncolytic Virotherapy	C-16
3.	Naaz Khan	AMP Feedback Loop in Psoriasis Dynamics: Bifurcation and Stability Analysis of a Th17-Th2-Keratinocyte-Cytokine Model	C-18
4.	Parthib Sengupta	A Unified Framework for Lyapunov Function Construction in Generalized Stochastic Population-Growth Models	C-19
5.	Gayatri Roy	Blockchain-Based Information Sharing and Transparency Decisions in CSR-Driven Supply Chains	C- 72
Parallel Session – C II -Mathematical Biology- Mathematical Modelling (05:30pm to 06:20pm)			
Venue: Technology Bhavan - Room No.: 503 -5th floor			
Chairperson: Prof. Fugo Takasu		Time: 10 minutes (each)	
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Sourav Maity	Harvesting Reshapes Resource Enrichment and Spatial Self-Organization in a Resource-Mediated Intraguild Predation System	C-21

7.	Arnabi Saha	A Fractional-Order Commensurate and Incommensurate Study on Predator-Prey System with Allee in Predator and Fear Effect	C-23
8.	Debabrata Mandal	Top-down regulation and catastrophic transition in an intraguild predation model driven by IG-prey's anti-predator nature	C-29
9.	Prajwal Gupta	Periodic Structures, Period-Bubbling, and Multistability in a Fractional-Order Ecological System with Allee Effect	C-30
10.	Sourav Ghorai	Turing Instability and Bistable Epidemic Spread on Complex Networks: Effects of Topology and Initial Seeding	C-12

Contributory Talk (04:40pm to 06:20pm)

Parallel Sessions (C I/C II/C III/C IV)

Parallel Session – C III– Theoretical Physics (04:40pm to 05:30pm)

Venue: Technology Bhavan - Room No.: 506 -5th floor

Chairperson: Dr. Nilanjana Mahata

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Anikul Islam	Exact Gravastar Solution	C-04
2.	Anisha Saha	On Conformal Fractional Ricci Flow	C-07
3.	Nilofer Ali	Study of Sasakian manifolds admitting hyperbolic Ricci soliton with respect to Zamkovoy connection	C-15
4.	Prosanta Mandal	Dynamics of particles around weakly magnetized Riessner Nordstrom black hole in presence of dark energy	C-17
5.	Titli Biswas	Study of Single-Step and Multistep Numerical Methods for the Two-Body Satellite Orbit Problem	C-44

Parallel Session – C III– Solid Mechanics-Fluid Dynamics (05:30pm to 06:20pm)

Venue: Technology Bhavan - Room No.: 506 -5th floor

Chairperson: Dr. Sanjib Kumar Paul

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
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6.	Zenith Oinam	Guided wave propagation under the influences of magnetic field and initial stress on in thin generalized magneto-thermoelastic plate with two temperatures	C-39
7.	Arambam Bickyrose Chanu	Reflection phenomena in a visco-thermo-micropolar elastic medium with the micro-temperature	C-57
8.	Lalruatpuui Colney	Propagation of Rayleigh wave in anisotropic thermo-piezoelectric Material	C-62
9.	H. Lalhriatpuia	Spillover-Free Finite Element Model Updating with Incomplete Spectral Data: An Optimization Approach	C-49
10.	Prasanta Mandal	Mixed Convective Heat Transfer and Entropy Generation in an Inclined Non-Square Cavity with a Power-Law Nanofluid	C-13

Contributory Talk (04:40pm to 06:20pm) Parallel Sessions (C I/C II/C III/C IV)			
Parallel Session – C IV- Mathematical Biology- Mathematical Modelling (04:40pm to 05:30pm) Venue: Technology Bhavan - Room No.: 507 -5th floor Chairperson: Dr. S. Alam Time: 10 minutes (each)			
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Anuj Kullu	Immune-Driven Evolutionary Escape in Breast Cancer: A Mathematical Framework for Immunoediting and Optimal Therapeutic Control	C-31
2.	Satyabhan Singh	Unveiling non-linear dynamics in the discretized Hastings-Powell system: A planar study of Arnold Tongues, bell-shaped periodic windows, and multistability	C-32
3.	Kanchan Singh	Qualitative analysis and Turing patterns in a prey-predator model incorporating Allee effect and anti-predator behavior	C-33
4.	Pinki Rani	Stability and Bifurcation Analysis of a Cancer–Immune Interaction Model under Chemotherapy	C-34

5.	Partha Biswas	The missing lever in Wolbachia-based mosquito control: female recruitment proportion outsmarts population replacement and population suppression	C-35
Parallel Session – C IV- Mathematical Biology- Mathematical Modelling (05:30pm to 06:20pm) Venue: Technology Bhavan - Room No.: 507- 5th floor Chairperson: Prof. Molla Basir Ahamed Time: 10 minutes (each)			
Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Mrinmay Pakhira	Trade-off between Allee remediation and disease transmission for adaptive survival: A mathematical model on Allee-exposed species	C-36
7.	Dr. Bapin Mondal	Stability and bifurcation analysis of an imprecise Bazykin's predator–prey model	C-37
8.	Animesh Manna	A Delay Differential Game for Blockchain-Enabled Sustainable Supply Chains with Recycling, Trade Credit and Green technology spillover	C-43
9.	Dipti Dolai	Analysis and Optimal Control of a Caputo Fractional-Order Ebola Model with Environmental Pathogen Transmission	C-46
10.	Ritika Chhetri	Mathematical and Computational Modelling of Harmful Algal Blooms: Critical Synthesis, Forecasting Challenges, and a Future Research Direction	C-50

Gala Dinner (6:30-9:00 PM)
Venue: University Guest House

End of Day 1

Day 2: 13th October, 2026

Plenary Talk 4 (10:20am-10:55am) Chairperson: Prof. František Štěpánek TEQIP Building (Room no: 208)		
Name: Prof. Stacey Smith? The University of Ottawa, Canada	Title of Talk: Modelling the Buruli Ulcer in Disparate Regions: Africa and Australia	Abstract Serial No. P-05
Plenary Talk 5 (10:55am-11:30am) Chairperson: Prof. Sujit Kumar Sardar TEQIP Building (Room no: 208)		
Name: Jaydeb Sarkar Indian Statistical Institute, Kolkata, India	Title of Talk: Approximation and interpolation	Abstract Serial No. P-12
Plenary Talk 6 (11:30am-12:05pm) Chairperson: Prof. Igor Schreiber TEQIP Building (Room no: 208)		
Name: Prof. David Greenhalgh University of Strathclyde, UK	Title of Talk: Modelling the effects of disease awareness programs amongst people who inject drugs.	Abstract Serial No. P-04
Tea Break 12.05am – 12.20pm		
Plenary Talk 7 (12:20pm-12:55pm) Chairperson: Prof. Sagnik Singha TEQIP Building (Room no: 208)		
Name: Prof. Sabyasachi Bhattacharya ISI, Kolkata	Title of Talk: Uncertainty as the Main Actor on Centre Stage in Resource Management and Disease Dynamics and across Ecology and Epidemiology	Abstract Serial No. P-13
Plenary Talk 8 (12:55pm-01:30pm) Chairperson: Prof. Bibhas Chandra Giri TEQIP Building (Room no: 208)		

Name: Prof. A. K. Mishra BHU, India	Title of Talk: Modelling the ability of mass drug administration to interrupt soil-transmitted helminth transmission	Abstract Serial No. P-01
<u>Lunch Break (University Guest House)</u> 01.30pm – 02.30pm		
Invited Talks (02:30pm to 04:00pm) Parallel Session-I		
Invited Talk 9(02:30pm-03:00pm) Chairperson: Prof. Sudeshna Banerjee Technology Bhavan (Room No: 502)		
Name: R. K. Mohanty South Asian University	Title of Talk: On the Bi-parametric Stability of High Accuracy ADI Method of O(2,4,4) for 2D Model Transmission Line Equation on an Unequal Mesh	Abstract Serial No. I-12
Invited Talk 10 (03:00pm-03:30pm) Chairperson: Prof. Anup Bandyopadhyay Technology Bhavan (Room No: 502)		
Name: Prof. Amit Kumar Ghosh Birbal Sahni Institute of Palaeoscience, India	Title of Talk: Application of Mathematics in Earth and Atmospheric Sciences	Abstract Serial No. I-07
Invited Talk 11 (3:30pm-4:00pm) Chairperson: Prof. Sukhendu Kar Technology Bhavan (Room No: 502)		
Name: Prof. Ravi Banavar Centre for Systems and Control, IIT Bombay, India	Title of Talk: Data Science, Geometry and Robust Control	Abstract Serial No. I-08
Invited Talks (02:30pm to 04:00pm) Parallel Session-II		
Invited Talk 12 (02:30pm-03:00pm) Chairperson: Prof. Shovonlal Roy Technology Bhavan (Room No: 503)		
Name: Prof. Sudipta De	Title of Talk: Mathematical Modeling Pathways for Optimal Sustainable Energy Systems	Abstract Serial No. I-14

Jadavpur University		
Invited Talk 13 (03:00pm-03:30pm) Chairperson: Prof. Hideki Miyachi Technology Bhavan (Room No: 503)		
Name: Dr. Debmalya Sain IIIT Raichur, India	Title of the talk: An improvement of the Blanco-Koldobsky-Turnšek characterization of isometries	Abstract Serial No: I-06
Invited Talk 14 (03:30pm-04:00pm) Chairperson: Prof. Sabyasachi Bhatyacharya Technology Bhavan (Room No: 503)		
Name: Dr. Sourav Rana IIST Shibpur, India	Title of the talk: Multi-region evaluation of a hybrid ARIMAX-ARNN model for dengue incidence forecasting	Abstract Serial No: I-13
Young Scientist Symposium (2:30pm-4:10pm) Parallel Session-III Technology Bhavan (Room No: 402)		
Chairperson: Prof. Joydeb Sarkar		Time: 20 minutes (each)
Name: Dr. Tanay Saha Mizoram University, India	Title of the talk: Distance to the Nearest Structured Polynomial Matrix with a Prescribed Eigenvalue Geometric Multiplicity	Abstract Serial No: S-02
Name: Dr. Ayan Ghosh NIT Durgapur, India	Title of the talk: Recent Advancements in the Theory of Characterized Subgroups	Abstract Serial No: S-06
Chairperson: Prof. Sukhendu Kar Time: 20 minutes (each)		

Name: Dr. Kalidas Mandal Jadavpur University, India	Title of Talk: On the preservation of parallel pairs and triangle-equality attainment pairs, with applications to isometries	Abstract Serial No. S-08
Name: Dr. Sandipan Dutta Mizoram University, India	Title of Talk: Equi-Baire One Property for $SL(3,H)$ actions on the Quaternionic Projective Plane	Abstract Serial No S-01
Name: Dr Abhishek Mukherjee Kalna College, India	Title of Talk: On singquandles arising from Core Quandles and their shadows	Abstract Serial No S-10

Tea Break (10 min)

Contributory Talk

**Contributory Talk (04:10pm to 05:50pm)
Parallel Sessions (C I/C II/C III)**

Parallel Session – C I- Mathematical Biology- Mathematical Modelling (04:10pm to 05:00pm)

Venue: Technology Bhavan - Room No.: 502- 5th floor

Chairperson: Prof. Humayan Kabir

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Raktim Ray	Forecasting Chaotic time-series using Nonlinear Observable Space Maps	C-51
2.	Shipan Chandra Deb nath	Clinical Burden of Dual Pathogens: A comparative analysis of Mono infection Versus Zoonotic co-infection outcomes using SIR method	C-52
3.	Prinsi Shukla	A nonlinear dynamical model of algal bloom regulation by herbivorous fish	C-54
4.	Anoop Kumar Mishra	Modeling the role of media coverage and recovery awareness programs on the dynamics of infectious diseases with limited healthcare facilities	C-55
5.	Vikrant Rai	A Mathematical Study of Continuous Model in Tumor–Immune Dynamics With Interleukin-2	C-61

Parallel Session – C I- Mathematical Biology- Mathematical Modelling (05:00pm to 05:50pm)

Venue: Technology Bhavan - Room No.: 502- 5th floor

Chairperson: Prof. Sabyasachi Bhattacharya

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Arijit Malakar	A comparative investigation of the delayed and non-delayed optimal control techniques on misinformation mitigation	C-70
7.	Rana Pal	Nonlinear Dynamics of Socio-Ecological Resilience and Mental Health: An Evolutionary Game Approach	C-71
8.	Kiriti Bhusan Mahato	Social dilemma analysis of an epidemic model with behavioral dynamics: An evolutionary game-theoretic approach	C-72
9.	Sagnik Bhaumik	Analysis of a non-autonomous dynamical system : Local stability and attractors	C-68
10.	Sanju Sardar	A Mathematical Model of Antidote-Induced Inhibition in Methanol Poisoning under Double Impulsive Therapy	C-69

Contributory Talk (04:10pm to 05:50pm)

Parallel Sessions (C I/C II/C III)

Parallel Session – C II- Fluid Dynamics- Mathematical Modelling (04:10pm to 05:00pm)

Venue: Technology Bhavan - Room No.: 503- 5th floor

Chairperson: Prof. R. K. Mohanty

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Sharif Mahmud Bhuiyan	Numerical Investigation of Fluid Mixing in Passive Microchannels with Varying Cross-Sections	C-24
2.	Ripa Muhury	Numerical Investigation of Mass Transfer Dynamics in Mango Slices during Osmotic Dehydration Using FEM	C-27
3.	Dhruba Majhi	Thermosolutal Convection of a radiative ternary hybrid nanofluid under an	C-45

		applied magnetic field in a porous cavity with half-sinusoidal heating	
4.	Asim Mahata	Electrophoresis of Polyelectrolyte-Adsorbed Soft Particles with pH-Regulated Charge Densities of the Inner Core and Shell	C-47
5.	Sounov Marick	Multiscale pattern complexity in dry-land vegetation: A simplified modeling approach incorporating root augmentation and stratified soil water dynamics	C-53

Parallel Session – C II- Fluid Dynamics- Mathematical Modelling (05:00pm to 05:50pm)

Venue: Technology Bhavan - Room No.: 503- 5th floor

Chairperson: Prof. Ujjwal Kumar Deb

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Gauri Agrawal	Modeling the role of aerial water spraying in combating atmospheric pollutants and sustain industrialization	C-56
7.	Jenny Lallawmzuali	Multi-Source BOD–DO Dispersion Model with Variable Hydraulics	C-59
8.	Sarna Soren	Double-diffusive convection of a Casson tri-hybrid nanofluid in a plus-shaped enclosure with a baffle	C-63
9.	K. Lalnunsiami	Adaptive Grid Refinement Framework For Air Quality Dispersion Modeling In Opencast Coal Mines	C-64
10.	Lal Prakash Baral	A Wind-Oriented Grid-Based Mathematical Framework for Efficient Source-Receptor Analysis of Particulate Dispersion in Open-Cast Coal Mines	C-65

Contributory Talk (04:10pm to 05:50pm)

Parallel Sessions (C I/C II/C III)

Parallel Session – C III- Operations Research (04:10pm to 05:00pm)

Venue: Technology Bhavan - Room No.: 506- 5th floor

Chairperson: Prof. Bibhash Chandra Giri

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Subhamoy Bera	Fairness and Power in Sustainable Supply Chain: A Differential Game with Blockchain-Enabled Coordination	C-05
2.	Pritam Majumdar	Blockchain-Based Information Sharing and Transparency Decisions in CSR-Driven Supply Chains	C-09
3.	Aditya Bag	Do Trade-Old-for-Cash and Trade-Old-for-New policies for end-of-life products benefit industries producing innovative eco-friendly products?	C-22
4.	Shantanu Mahato	Optimal Replenishment and Shipment Decisions for a Manufacturer–Retailer System under Trade Credit and Cost Uncertainty	C-42
5.	Shruti Mahato	Game-theoretic framework for centralised and decentralised supply chain coordination under imperfect production and trade credit	C-60

Parallel Session – C III- Modelling and interdisciplinary Science(05:00pm to 05:50pm)

Venue: Technology Bhavan - Room No.: 506- 5th floor

Chairperson: Prof. Tapan Kumar Kar

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
6.	Nivedita Singh	Models on Tramadol and Polysubstance Use	C-48
7.	Avishek Chowdhury	Quantifying Explanation Shift in Deep Neural Networks: A Mathematical Modelling Framework for Adversarial Detection	C-26
8.	Oindrella Dey	Analyzing Control Strategies in DC/DC Buck Converters: A Mathematical and Simulink-Based Approach	C-28

|| Cultural Evening ||

**Venue: Dr. Triguna Sen Auditorium, Jadavpur University, Kolkata,
Time: 5:30 PM**

- **INAUGURAL SESSION:**

“Bhorbo Bhuban Gaaner Sure”

- **Students of Faculty of Science, Jadavpur University**

‘Bhorbo Bhuban Gaaner Sure’ is the tribute to the Culture of Bengal, where we worship The Goddess Durga by our voice of joy. In this joy our heart desires to fill the world with the melody of songs. "O Mother! Bless us by your divine powers. We bow to You.

- **RECITATION:**

"Transcending the Existential Crisis: A Quest for Consciousness and Unity"

- **Students of Faculty of Science, Jadavpur University**

The core theme of this poetic collage explores the deep textures of human existence—the relentless flow of time, the profound loneliness of the modern soul, and the crises of identity. Yet, it moves beyond despair, culminating in a powerful awakening of the collective human spirit, intellect, and courage to build a fear-free tomorrow.

- **CLASSICAL DANCE BATTLE:**

“Katthak vs Bharatnatyam”

- **Gunja Mukherjee and Mahika Ghosh, Department of Mathematics, Jadavpur University**

Two classical traditions, two powerful stories, one stage. From the temples of South India comes Bharatanatyam, rooted in precision, expression and grace. From the storytelling traditions of North India comes Kathak, born from the Kathakas who brought tales to life through rhythm and movement. Tonight Bharatanatyam’s strength meets Kathak’s rhythm, footwork and dazzling spins. Let the ghungroos speak, let the beats collide, and let the classical battle begin!

- **Mono Drama:**

“Clinging to Gitanjali: A Journey from Grief to Light (গীতাঞ্জলিকে আঁকড়ে ধরে)”

- Written and Performed by Prof. (Dr.) Joydev Chattopadhyay, Indian Statistical Institute, Kolkata

Synopsis

Set against the backdrop of quiet introspection, *Clinging to Gitanjali* weaves an extraordinary tapestry connecting two distant souls across time and geography. The narrative begins in the trenches of the First World War with young British war poet Wilfred Owen, whose parting words to his mother were drawn from Rabindranath Tagore's *Gitanjali*. Through this profound cross-cultural encounter, the performer traverses Tagore's own landscape of devastating personal bereavements—the loss of his mother, sister-in-law Kadambari Devi, wife Mrinalini, daughter Renuka, and beloved youngest son Shamindranath. It is a contemplative journey demonstrating how unendurable human sorrow transfigures into universal light, sublime surrender, and spiritual resilience.

The Historic Letter: A Mother's Solace

In August 1920, Susan Owen wrote to Rabindranath Tagore, recalling her son Wilfred Owen's final departure for the front lines in 1918 before he was killed in action just a week before the Armistice:

"Dear Sir Rabindranath, ...Nearly two years ago, that dear boy went out to the war for the last time and the day he said Goodbye to me—we were looking together across the sun-glorified sea towards France, with breaking hearts—when he, looking up, repeated to me the wonderful words of yours:

'When I go from hence let this be my parting word, that what I have seen is unsurpassable.'

...I have been trying to find the whole poem itself in your books—but I have not been able to. Would you tell me where it is to be found?" (Song 96, Gitanjali / Song Offerings)

Philosophical Essence of the Performance

- **Grief Transmuted into Light:** Rather than succumbing to existential despair, Tagore embraced bereavement as a gateway to understanding life's unbroken continuum—where "being" and "non-being" (আছে and নাই) merge into the same infinite ocean.
- **The Solitary Isle of Devotion:** Unlike his other works, Tagore dedicated *Gitanjali* to no mortal; it was his intimate sanctuary—a bridge between personal anguish and absolute surrender to the '*Nibhrito Praner Debata*' (the Deity of the Secret Soul).
- **The Divine in the Storm:** Transcendence is discovered not in passive serenity, but by walking fearlessly into the tempests of existence. When every outer light is extinguished, the inner song awakens.

(Note for international delegates: The performance is rendered in Bengali with synchronised visual projections and thematic English subtitles.)

- **Dance Drama:**

“Mohishasur Mardini”

- **Students of Faculty of Science, Jadavpur University**

Mahishasura Mardini is a vibrant theatrical portrayal of the divine emergence and triumph of **Devi Durga**, celebrating her as the embodiment of courage, Shakti (Power), and divine grace. Through a harmonious blend of classical dance, expressive movement, and dramatic dialect, the performance brings alive the timeless battle between Durga and Mahishasura, while subtly reflecting her enduring influence as a symbol of hope, strength, and resilience.

- **Group Song:**

“Epitome of the soul”

- **Students of Faculty of Science, Jadavpur University**

Whether it is the radiance of tradition, the fervour of religious emotion, the cherished memories of friendship, or the pain of separation—beyond the boundaries of caste and creed, the religion of humanity becomes the final hymn, while the dignity of motherhood becomes the ultimate ode. Across the seven colours of the human heart unfolds an extraordinary and novel fusion of love, nostalgia, spirituality, and nationalism.

- **SPEECH:**

“If Mathematics Is Universal, Why Isn't It Accessible?”

-Naaz Khan and Parthib Sengupta, Department of Mathematics, Jadavpur University

Mathematics speaks the same truth to everyone. But does everyone get to hear it? Behind the idea of mathematics as a universal language lies unequal schools, costly coaching, language barriers, limited resources, and examinations that can turn a shared human heritage into a barrier. Mathematics itself knows no caste, race, wealth, privilege. It belongs to everyone. Yet access to its learning and understanding remains deeply unequal. This speech is not an argument against mathematics, but for reclaiming it as a tool of freedom- the freedom to question, think, reason, and imagine. In the age of Artificial Intelligence, when machines can increasingly provide mathematical answers, this freedom to question becomes even more important. Mathematics belongs to everyone. But does everyone have an equal opportunity to belong to mathematics?

- **Dance Drama:**

“CHITRANGADA”

- Students of Faculty of Science, Jadavpur University

Rabindranath Tagore's Chitrangada is a powerful exploration of identity, womanhood, and the pursuit of self-worth beyond conventional notions of beauty and femininity. Drawing from the Mahabharata, the work portrays Chitrangada, a warrior princess whose strength, courage, and leadership challenge the traditional boundaries imposed upon women. Her encounter with Arjuna brings forth an inner conflict between her authentic self and the socially celebrated ideal of feminine beauty. Through this transformation, Tagore questions whether external appearance can ever define the completeness of an individual. The narrative ultimately moves beyond the desire for acceptance towards self-realisation and dignity. Chitrangada thus remains deeply relevant in the contemporary world, where questions of identity, gender expectations, and individuality continue to shape human experience.

End of Day 2

Day 3: 14th October, 2026

Plenary Talk 9 (10:00am-10:35am) Chairperson: Prof. Shamik Ghosh TEQIP Building (Room No: 208)		
Name: Prof. Arup Bose ISI, Kolkata	Title of Talk: Introduction to random matrices and non-commutative probability	Abstract Serial No. P-02
Plenary Talk 10 (10:35am-11:10am) Chairperson: Prof. David greenhalgh TEQIP Building (Room No: 208)		
Name: Prof. Robert Haki Institute of Mathematics, Czech Academy of Sciences, Czech Republic	Title of Talk: Periodic Solutions of Singular Equations: Old Questions and New Results	Abstract Serial No. P-07
Plenary Talk 11 (11:10am-11:45am) Chairperson: Prof. Shyamal Dana TEQIP Building (Room No: 208)		
Name: Prof. Shovonlal Roy University of Reading, UK	Title of Talk: Satellite-Guided Individual-Based Modelling for Mechanistic Prediction: Insights from Marine Fish Populations	Abstract Serial No. P-11
Plenary Talk 12 (11:10am-11:45am) Chairperson: Prof. Pratulananda Das TEQIP Building (Room No: 208)		
Name: Prof. Hideki Miyechi School of Mathematics and Physics, Japan	Title of Talk: Deformation space of Riemann surfaces: Analysis and Geometry	Abstract Serial No. P-10
Plenary Talk 13 (11:10am-11:45am) Chairperson: Prof. A.K. Misra TEQIP Building (Room No: 208)		
Name: Prof. Humayan Kabir Jahangirnagar University, Bangladesh	Title of Talk: Modeling the Spread of Neolithic Agriculture through Go-or-Grow Dynamics	Abstract Serial No. P-14

Contributory Talk

Contributory Talk (02:30pm to 03:20pm)	
Parallel Sessions (C I/C II)	
Parallel Session-CI - Mathematical Biology- Mathematical Modelling- (02:30pm to 03:20pm)	
Venue: Technology Bhavan - Room No.: 502- 5th floor	
Chairperson: Prof. Debashis Mukherjee	Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	SK Mosaraf Ahammed, Kunal Paul	Dynamical Analysis of Sterile Insect Technique for Sustainable Pest Control in Jatropha Plantations	C-11
2.	Homagnic Bose	Insight of Vector's Feeding Affinity in Zoonotic Cutaneous Leishmaniasis Transmission: A Case Study	C-75
3.	Tarun Mondal	Adaptive Contact Behavior under Awareness Campaigns: A Case Study of an Epi-Economic Model of Leprosy	C-20
4.	Baishaki Tikader	Do Trade-Old-for-Cash and Trade-Old-for-New policies for end-of-life products benefit industries producing innovative eco-friendly products?	C-22
5.	Satyajit Mukherjee	An Adaptive Hybrid Framework for Spatiotemporal Forecasting and Risk Assessment of Cutaneous Leishmaniasis in Iran	C-76

Parallel Session – C II- Differential Geometry and Operator theory (02:30pm to 03:20pm)

Venue: Technology Bhavan - Room No.: 503- 5th floor

Chairperson: Prof. Amit Kumar Ghosh

Time: 10 minutes (each)

Serial Number	Speaker	Title of the Talk	Abstract Serial No.
1.	Somdatta Barik	Weyl-type operators on weighted Bergman spaces: Berezin range and Berezin number	C-74
2.	Laldinpuia	Certified Stability Radii and Variance-Based Sensitivity of a Fuzzy AHP Soil Health Index: Analytical Results and Validation on Acidic Hill Soils of Mizoram, India	C-70
3.	Sandip Mistry	Diffraction of Love Waves by a Rigid Strip at the Interface of an Orthotropic Layer and an Isotropic Half-Space	C-66
4.	Dr. Subhash Chandra Mondal	Magnetoelastic Impact Response of a Rigid Disk at an Isotropic–Anisotropic Interface	C-77
5.	Rahul Mandal	An Overview of Complex Kleinian Groups	C-78

Poster Presentation

Venue: Technology Bhavan - Room No.: 506- 5th floor

Time: 2:30-3:00 PM

Serial Number	Presenter	Title of the Presentation	Abstract Serial No.
1.	SHUBAM	Model Discovery in Structurally	PP-1

	KANDYAL	Sensitive Nonlinear Systems Using a Data-Driven Approach	
2.	Narendranath Mitra	Impossibility in Constructing 1 Radian Explained - An Approach to Better Approximation	PP-2
3.	Ramotjaki Lucky Monaledi	Impact of Nanoparticles Dispersion on Entropy Generation in a Microchannel Poiseuille Flows: A Mathematical Model	PP-3
4.	Shubham Mondal	Biomathematics in Cancer Research: Looking into how Mathematical Models Are Used to Understand Tumor Growth and the Effectiveness of Different Treatment Strategies	PP-4

VALEDICTORY SESSION

Announcement of

- ❖ Prof. A.B. Roy Lifetime Achievements Award
- ❖ Top 3 Best Contributory Speaker Awards
- ❖ Best Speaker Award at ‘Young Scientist Symposium’

Keynote/Plenary/Invited Speeches

❖ Applications in Mathematics (Applied Mathematics):

1. Applications of mathematical modelling in pharmaceutical product and process development

[K-01]

František Štěpánek

Department of Chemical Engineering, UCT Prague, Czech Republic

Abstract: The development of new pharmaceutical products and their manufacturing processes requires a series of complex decisions in a multi-dimensional parametric landscape. These range from decisions on the solid-state form of the active drug substance (amorphous vs. crystalline state, particle size distribution), composition of the dosage form (grade and percentage of excipients such as fillers, binders, wetting agents, stabilisers, glidants, disintegrants), and the manufacturing process route (milling, sieving, granulation, blending, tableting, coating). To support these decisions, mathematical modelling combined with experimental data plays a crucial role. The talk will cover examples of industrially relevant formulation and processing steps in which recent advancement in both methodology and computing power enabled the incorporation of mathematical modelling as an integral part of the R&D workflow. These include the formation of pharmaceutical nano-suspensions by wet stirred media milling and by continuous liquid antisolvent precipitation, and the design of long-acting injectables with respect to the effect of particle size distribution on the pharmacokinetic profile of the active substance. The talk will conclude with an outlook for the future needs and opportunities in the field of mathematical modelling in pharmaceutical product and process development.

2. Modelling of metabolic oscillations in cyanobacteria

[P-06]

Igor Schreiber

University of Chemical Technology, Prague

Abstract: Cyanobacteria rely on using two metabolic processes: photosynthesis to produce energy-rich carbohydrates, and fixation of nitrogen to produce nucleic and amino acids. These processes are mutually inhibiting and lead to experimentally observed time-separated oscillatory behavior in a population of cells, even under constant intensity of light when the cells are still viable and display oscillations measured by their oxygen release. Such oscillations are assumed to be based solely on metabolism, have period much shorter than the circadian rhythms (about 12-17 h) and have been termed ultradian cycles. Such temporal patterns are a natural way for the cells to survive. We use the theory of reaction networks to construct a compartmental model of metabolic oscillatory dynamics observed in experiments

with photosynthesizing diazotrophic cyanobacteria in a flow-through photobioreactor. The model describes dynamics of component concentrations in two parts of the cell suspension - solution and cellular biomass. Components in the solution are nutrients (CO_2 , N_2 , O_2), components in the biomass are represented by functional compartments that interact via power law chemical kinetics. The compartments include nutrients inside the cells and specific as well as generic functional/structural compartments. The model utilizes experimentally determined biomass growth rate expressed in terms of the Monod kinetics. The crucial part of modelling is determination of rate coefficients in the power law expressions. We employ the method of constrained stoichiometric analysis, which is a combination of stoichiometric network theory and convex linear optimization. The method has been developed in our previous work and successfully used to determine unknown kinetic parameters in oscillatory enzyme reactions. The model is then used to simulate ultradian oscillations under experimental conditions and provides results closely agreeing with the experimentally observed 14-hour ultradian cycles.

3. Modelling the Buruli Ulcer in Disparate Regions: Africa and Australia

[P-05]

Stacey Smith?

The University of Ottawa, Canada

Abstract: The Buruli ulcer (BU) is a neglected skin necrotising disease caused by *Mycobacterium ulcerans* that is endemic in tropical regions, particularly in Africa. Recently, BU has spread to Australia, where the modes of transmission have adapted to local animal hosts and vectors. We first perform a statistical analysis of Buruli ulcer data in order to inform our model development. We then formulate mathematical models for both Africa and Australia that incorporate seasonality, ulcerative and non-ulcerative cases, treatment methods and multiple environmental factors. We model the change of transmission mode from aquatic and environmental transmission in Africa to mosquito and environmental transmission in Australia and apply a cost-optimisation framework to determine which human interventions can be most effective. We determine conditions for the stability of the disease-free equilibrium and prove that the system undergoes a transcritical bifurcation at $R_0=1$. A cost-optimisation analysis shows that education and vector control are the most cost-effective interventions. With the recent relocation of the Buruli ulcer to new environments, public-health interventions should be calibrated for local transmission routes.

4. Modelling the effects of disease awareness programs amongst people who inject drugs

[P-04]

David Greenhalgh

University of Strathclyde, United Kingdom

Abstract: Mathematical modelling techniques have been used extensively during the HIV epidemic. We develop a mathematical model of the effect of disease awareness programs on

the prevalence of HIV amongst people who inject drugs (PWIDs) building on models developed previously. A system of differential equations have been developed which introduces the disease awareness program by reducing the disease transmission factor by a simple multiplicative factor. We perform an equilibrium and stability analysis. The key biological parameter is the basic reproduction number R_0 . If $R_0 \leq 1$ then the disease free equilibrium (DFE) is the unique equilibrium. In this case the DFE is globally asymptotically stable. If $R_0 > 1$ then there is the DFE and additionally a unique endemic equilibrium. If there is initially disease present and $R_0 > 1$ then the system tends to the unique endemic equilibrium. The DFE is locally asymptotically stable if $R_0 < 1$, neutrally stable if $R_0 = 1$ and unstable if $R_0 > 1$. If $R_0 > 1$ we showed that the endemic equilibrium was locally asymptotically stable too. Our analytical results are confirmed by numerical simulations in MATLAB using realistic parameter values. An application to hepatitis C virus will be also discussed.

5. Data Science, Geometry and Robust Control

Ravi Banavar

Centre for Systems and Control, IIT Bombay, India

Abstract: Data Science is playing an increasing role in many engineering applications. This talk will focus on the geometric aspects of data-driven control. Data is almost, always corrupted. Any least-squares interpretation is hence subject to this uncertainty in the subspace on which we obtain meaningful information. We model this "uncertainty" as a metric ball on the Grassmannian manifold, leading to a min-max problem over Euclidean and manifold variables. The inner maximization admits a closed-form solution, enabling an efficient algorithm with a transparent geometric interpretation. Applied to robust finite-horizon linear-quadratic tracking in data-enabled predictive control, the method improves upon existing robust least-squares formulations, achieving stronger robustness and favourable scaling under small uncertainty.

6. Mathematical modeling and analysis of the impacts of climate change, human population, and development activities on agricultural crop yield

A. K. Mishra

[P-01]

Department of Mathematics, Institute of Science, Banaras Hindu University, Varanasi, India.

Abstract: At present time, sustainable crop production and environmental stability are of prime importance due to the unprecedented expansion of the human population and the escalating threats of human-driven climate change. The rising atmospheric carbon dioxide (CO_2) level and the subsequent elevation in the global average surface temperature pose some of the most significant challenges to global food security. While a moderate increase in CO_2 and temperature may initially boost crop growth through enhanced photosynthesis, exceeding critical thermal thresholds can disrupt plant physiology, reduce leaf area, and

severely impair overall agricultural productivity. Furthermore, climate change intensifies biotic threats, such as frequent insect outbreaks, which adversely affect crop health. Simultaneously, rapid urbanization and development activities diminish cultivable land and exacerbate greenhouse gas emissions. Mathematical models based on the complex interactions among greenhouse gas emissions, temperature variations, human activities, and crop yields provide a robust foundation to systematically analyze and mitigate the adverse effects of climate change on agriculture. These models' analysis offers a systematic approach to understand the dynamics involved, enabling informed decisions for the promotion of sustainable and secure crop production amid the escalating challenges.

In this talk, we will comprehensively explore these dynamics through a series of mathematical models. First, we will discuss a nonlinear mathematical model to investigate the combined effects of elevated CO₂ and temperature on crop yield. Our model considers that surface temperature initially boosts crop growth until a critical optimal threshold is reached, after which the growth rate rapidly declines. We will also explore seasonal variations, to demonstrate how crops with higher temperature tolerance can yield better even under elevated CO₂ conditions. Since rising temperatures not only impact crops directly but also accelerate the metabolism and reproduction of pests, it is crucial to analyze their joint ecological effects. Therefore, we will analyze a mathematical model to understand the consequences of elevated temperature on both crop yield and insect population dynamics. We will illustrate how higher temperatures escalate insect population growth, which negatively impacts crop production, and how the system exhibits various bifurcations depending on crop consumption rates and environmental thresholds. Due to the continuous expansion of the human population, agricultural lands are increasingly encroached upon for residential and commercial purposes, further amplifying CO₂ emissions and reducing the carrying capacity of crops. Therefore, in the next part, we will discuss a mathematical model unraveling the combined effects of increasing human population and urbanization on CO₂ levels and crop yield, utilizing both deterministic and stochastic modeling approaches. We will highlight critical thresholds in anthropogenic CO₂ emissions, beyond which crop productivity destabilizes. Finally, since the depletion of forest density due to development activities is a primary driver of elevated CO₂ levels, we recognize the need for an innovative mitigation strategy. In the last, we will analyze a non-linear mathematical model concerning the strategy of maintaining the atmospheric level of CO₂ alongside ongoing development activities. This novel approach evaluates the specific proportion of deforested land that must be allocated for the planned plantation of leafy trees, which absorb significantly more CO₂ than regular trees to counterbalance emission impacts. This research determines critical emission thresholds, optimal temperature ranges, and viable land-use strategies for safeguarding agricultural productivity and environmental stability in the face of ongoing climate change.

7. Satellite-Guided Individual-Based Modelling for Mechanistic Prediction: Insights from Marine Fish Populations

[P-10]

Sovonlal Roy

University of Reading, United Kingdom

Abstract: Populations emerge from complex interactions between environmental conditions, individual physiology, behaviour and population processes. Many conventional population

dynamics models, however, represent these processes implicitly, making it difficult to link environmental variability to individual responses and emergent population dynamics. This talk presents a modelling framework that integrates large-scale environmental observations with mechanistic representations of individual growth, energy allocation, movement and survival. The framework uses satellite-derived environmental and ecological information to provide spatially and temporally varying conditions experienced by individuals. The model simulates individual dynamics in response to local environmental conditions and food availability, allowing population structure and dynamics to emerge from the collective behaviour of individuals. Marine fish populations provide examples of how this approach can be used to reproduce observed population dynamics, predict spatial distributions and investigate responses to changing environmental conditions. Using specific examples of pelagic fish, including Atlantic mackerel and European seabass, the presentation will demonstrate how alternative formulations of movement, habitat selection and interactions among individuals can lead to different emergent spatio-temporal distributions and long-term population dynamics. Examples will also illustrate how model outcomes could inform management decisions for sustainable fisheries. Some future directions of such observation-constrained modelling frameworks will also be discussed in the context of population forecasting and assessment of environmental change.

8. Periodic Solutions of Singular Equations: Old Questions and New Results

[P-07]

Robert Hakl

Institute of Mathematics, Czech Academy of Sciences, Czech Republic

Abstract: Second-order differential equations with singularities in the phase variable arise naturally in a variety of physical models, ranging from celestial mechanics and bubble dynamics to nonlinear optics and wave propagation. At the same time, the presence of a singularity makes even the basic question of existence of periodic solutions substantially more delicate.

Starting from the classical results of Lazer and Solimini, we will give a brief overview of the development of this theory, with particular attention to attractive, repulsive, and indefinite singularities. We will illustrate how the existence and multiplicity of periodic solutions depend on a subtle interplay between the strength of the singularity, the regularity of the forcing term, and its sign-changing structure. Some sharp results and counterexamples will be discussed, as well as equations with competing or multiple singularities.

The talk will also highlight several applications that have motivated these questions and indicate some recent connections with periodic travelling waves in inhomogeneous media.

9. Point pattern approach to study spatial population dynamics

TAKASU Fugo

Nara Women's University, Japan

Abstract: In this talk, I focus on the "point pattern approach" where individuals are represented as points on continuous space and each point gives birth, dies, moves and

changes its status with a certain rule. This approach allows us to implement mechanistic interactions at individual level as an individual-based model. A point pattern can be quantified by focusing on the number of points (or singlets) as the 1st order structure, pairs made by two points displaced by a certain vector as the 2nd order structure, and triplets made by three points as the 3rd order structure, etc. Law and Dieckmann and Dieckmann and Law (2000) propose the method of moments that translates a set of individual-based algorithms to the dynamics of the 1st moment as the density of points N and the 2nd moment as the density of pairs $C(\xi)$ displaced by a vector ξ . The method of moments can be extended to handle "status change" of each point, e.g., Susceptible to Infectious to Removed, etc. Analysis of the derived moment dynamics as integro-differential equations is a key issue. I give several applications of the point pattern approach in spatial population dynamics and discuss potential future research based on point process.

10.TopoSSR: A Topological Data Analysis Framework for Identifying Disease-Associated Protein Structural Mutations

Amit Chakraborty

Sikkim University, India

Abstract: Identifying how disease-associated mutations affect protein structure remains a major challenge because many current computational approaches rely on sequence conservation, coordinate alignment, or global energetic measures, which fail to capture localized conformational changes within secondary structural elements. We introduce TopoSSR, a Topological Data Analysis (TDA) framework that detects mutation-induced structural deformations via persistent homology. The approach divides proteins into secondary structural blocks, depicts each block using a nine dimensional internal torsion-angle feature space, and employs Gaussian process based densification to maintain local geometric continuity. Persistent homology is calculated separately for each block, and the suggested Maximum Persistent Betti-1 (MaxPB-1) descriptor is utilized to assess topological complexity and detect structurally changed areas in wild-type and mutant proteins. Unlike traditional coordinate-based comparisons, TopoSSR is rotation- and translation-invariant, avoiding explicit structural alignment by relying on intrinsic geometric properties. The framework was tested on 10 experimentally established wild-type/mutant protein pairings, including viral proteins, tumor suppressors, metabolic enzymes, signaling proteins, and transport proteins. Across these different systems, TopoSSR reliably localized mutation-associated structural perturbations to distinct secondary structure blocks by MaxPB-1 characterization. The most dramatic deformations occurred in β -sheets and loop regions, whereas α -helices remained relatively stable. Case studies using the PIN1 isomerase and the SARS-CoV-2 Omicron spike protein highlight the framework's capacity to detect mutation-related structural reorganizations. These results establish TopoSSR as a scalable and theoretically robust framework for localizing mutation-driven structural changes. They also give a novel topological perspective for protein variant interpretation.

11. Application of Mathematics in Earth and Atmospheric Sciences

Amit Kumar Ghosh

Birbal Sahni Institute of Palaeosciences (Under Department of Science & Technology, Government of India)

Abstract: Geomathematics is one of the specific offshoots of Earth Sciences that is focussed in solving the problems related to various aspects of geosciences namely, geophysics, hydrology, geodynamics and seismology mineral and hydrocarbon exploration, evolution, paleoclimatology, paleontology, sedimentology and a wide array of applications using mathematical methods as tools. Basically, Geomathematics or Mathematical Geosciences is an interdisciplinary field. Different methods of mathematics solve complex problems in the Earth sciences. This interdisciplinary field dynamically fill the gap of knowledge between theoretical mathematics and practical geoscience applications that enables advanced modelling and simulation of geological processes. With strong mathematical foundations of Earth's systems, earth scientists contribute significantly to predict natural disasters, exploration of natural resources, and interpretation on climate change impacts. Mathematical frameworks using differential equations and fluid dynamics can develop general circulation models that are very useful for the prediction of changes in climate patterns and assessment of potential impacts. Magnitude of earthquake are quantified logarithmically using mathematical scales i.e., Richter scale. Differential equation e.g., Darcy's law for fluid flow in soil and statistical methods are utilized for the estimation of subsurface hydrocarbon reservoirs, ground water levels, and mineral reserves. In disaster management numerical simulations are helpful in forecasting the volcanic eruptions, tsunamis, floods, and landslides. Mathematical Geosciences also can measure meandering and erosion of rivers and landscapes, melting and retreat of glaciers alongside sea-level fluctuations. By analyzing geophysical data Geophysical Inverse Theory, we can measure the Earth's interior from the measurement of the earth's surface. In the classical areas of Geosciences like paleontology, evolution, crystallography and sedimentology use mathematics is well established. In paleontology, specifically in micropaleontology mathematics is applied as a tool for morphometric analysis of microfossils. Mathematical modelling was used around three decades ago on Silicoflagellates (silicified marine planktonic alga) skeletal morphology and a computer programme for three dimensional measurements of silicoflagellate skeletons was established. For the evolutionary perspective mathematics are widely used in cladistic taxonomy which is method of biological classification that clubs species into categories on the basis of shared evolutionary ancestry instead of overall physical similarities. CLAMP (Climate Leaf Analysis Multivariate Program) is very useful in decoding climatic signals archived within the physiognomy of fossilized leaves and it can effectively quantify past terrestrial climatic condition. Though in sedimentology mathematical application is used for determining the sedimentation rate, End-Member Modeling Analysis (EMMA) is rather a recent statistical method that determines unmixing multimodal grain-size distributions and identifies and quantifies processes of generation of sediments, transportation as well as deposition. For the validation of grain-size analysis of the sediments End-Member algorithms are extensively employed in Earth Sciences. So, it is evident that there are multifaceted applications of mathematics in earth sciences and there are ample scopes of validation of the datasets generated by earth scientists using applied mathematics.

12. Uncertainty as the Main Actor on Centre Stage in Resource Management and Disease Dynamics and across Ecology and Epidemiology

[P-12]

Sabyasachi Bhattacharya

Agricultural & Ecological unit, Indian Statistical Institute, Kolkata, India

Abstract: Uncertainty serves as a vital yardstick in both resource management and disease dynamics, fundamentally shaping outcomes in ecology and epidemiology. In this talk, uncertainty is viewed under two broad and interconnected spectrums: ecological uncertainty emerging from environmental variability, and epidemiological uncertainty arising from infectious disease processes, contamination, and population-level feedbacks. The focus is on integrating these two aspects to better understand extinction risk and sustainability in natural systems. From an ecological resource-management perspective, we employ Itô's lemma and first-passage-time probability to derive a critical threshold of environmental contamination, crossing which a population converges almost surely either to its carrying capacity or to the Allee threshold with quantifiable probabilities. Using Canine Distemper Virus in African wild dogs as a motivating epidemiological system, we introduce the **Disease–Stochasticity–Tipping (DST) triangle**, a novel framework developed within a stochastic SIR setting that captures vulnerabilities arising from the interaction of disease resilience, stochastic fluctuations, and tipping phenomena, particularly in the presence of Allee effects. This framework highlights how uncertainty can drive populations toward recovery or collapse even under similar management interventions. Furthermore, theoretical bounds on environmental noise intensity are used to estimate the present and future status of species under stochastic forcing. The applicability of the proposed framework is illustrated using large abundance data from the Global Population Dynamics Database (GPDD) across four taxonomic groups—birds, insects, mammals, and fish. Collectively, this work provides a unified epidemiological–ecological framework for guiding disease management and conservation strategies under pervasive uncertainty.

13. Lamb wave in the plate of microstretch thermoelastic diffusion materials

S Sarat Singh

Department of Mathematics and Computer Science, Mizoram University, India

Abstract: The present paper investigates the three thermoelastic theories on the propagation of Lamb wave in a linearly isotropic microstretch diffusion plate subjected to the thermally insulated/impermeable and isothermal/isoconcentrated boundary conditions. The secular equations of the Lamb wave are obtained for both symmetric and anti-symmetric modes of vibration. The secular equations for Rayleigh surface wave at short wavelength and plate wave at longer wavelength are obtained for both the boundary conditions from symmetric vibration. We also obtain the secular equation of flexural wave from the anti-symmetric vibration at the longer wavelength compared with the thickness of the plates. The phase velocity and attenuation are computed numerically for a particular model, and these results are compared for the coupled thermoelasticity(CT), Lord–Shulman and Green–Lindsay theories. There are three modes of dispersion and attenuation for each symmetric and anti-symmetric vibration. The dispersion curves of the Lamb wave increase from first to third mode of symmetric vibration in both thermally insulated/impermeable and isothermal/isoconcentrated plates. Certain special cases are reduced from the current formulation.

14. Quantification of Runoff and Sediment Transport for Sustainable Reservoir Management Using a Computational Modelling Approach

M. SUNDARARAJAN

Department of Mathematics and Computer Science, Mizoram University, India

Abstract: Recent satellite-based assessments indicate that approximately 25% of the storage capacity of Indian reservoirs has already been lost, primarily due to sediment accumulation. With rapid population growth and expanding urban and industrial development, cities and townships increasingly depend on municipal water supplies, making surface-water reservoirs critical for meeting water demands. The situation becomes particularly severe during summer, when many regions of India experience acute water shortages. This research investigates reservoir sedimentation resulting from soil erosion and sediment transport within river basins. It presents a computational approach for processing and managing large volumes of spatial and hydrological data associated with watersheds, river reaches, runoff, and sediment movement through streams, tributaries, and rivers into reservoirs. A computer algorithm has been developed to transform geographical watershed information into matrix-based structures, enabling efficient management of multiple river attributes while optimizing memory utilization. An appropriate sediment-transport model is applied to estimate runoff and associated sediment loads entering reservoir systems. Based on the modelling results, the study further proposes strategies for sustainable reservoir and water-resource management, with the objective of reducing sedimentation, preserving storage capacity, and improving the long-term reliability of surface-water resources.

14. Mathematical Modeling Pathways for Optimal Sustainable Energy Systems

Sudipta De

Department of Mechanical Engineering, Jadavpur University, India.

Abstract: Sustainable development has emerged as a global priority with seventeen Sustainable Development Goals (SDGs) established by the United Nations. Although the objectives of sustainable development are widely recognized, identifying viable and sustainable solutions for complex real-world systems remains challenging. Such solutions are inherently interdisciplinary and typically require the simultaneous consideration of multiple, and sometimes competing, criteria. Consequently, there is a need for systematic methodologies that can translate the broad objectives of sustainability into measurable criteria and provide a rational basis for decisionmaking. The energy sector is particularly critical to sustainable development. Developing robust methodologies for designing and evaluating future sustainable energy systems is therefore essential for researchers, planners, and policymakers. Mathematical modeling provides a cost-effective and structured approach for representing alternative energy systems, predicting their performance, and exploring different technological and operational scenarios. When integrated with appropriate mathematical and analytical tools for sustainability assessment, such models can facilitate the systematic evaluation of multiple sustainability criteria and the identification of suitable solutions from a range of alternatives. This presentation will discuss a mathematical modeling pathway that integrates energy-system modeling with multi-criteria sustainability assessment. The approach combines domain-specific knowledge of energy systems with mathematical and analytical techniques to quantify and evaluate different dimensions of sustainability. The fundamentals of sustainability and the associated challenges in translating its broad objectives

into measurable assessment criteria will be outlined. A mathematical framework for sustainability assessment developed by our research group will then be presented. Finally, selected case studies based on energy-system models and real-world data will be discussed to demonstrate this framework.

15. Effects of Solar Related Temperature Variations on Microalgae Cell Growth in a Photobioreactor

Ujjwal Kumar Deb

Chittagong University of Engineering & Technology, Bangladesh

Abstract: Temperature is an important factor affecting microalgal growth, particularly in outdoor cultivation systems where environmental conditions vary throughout the day. In this study, a computational fluid dynamics (CFD) model was developed by coupling Laminar Two-Phase Flow for carbon dioxide and microalgal suspension with Heat Transfer in Fluids and Transport of Diluted Species. The model was used to investigate temperature distribution and hydrodynamic behavior in a Horizontal Loop Tubular Photobioreactor (HLTP) exposed to solar radiation. Radial and axial temperature variations were investigated at different reactor locations and times of the day. The radial temperature profiles showed higher temperatures near the tube wall and lower temperatures toward the reactor center. In the U-loop, an asymmetric temperature distribution was observed due to enhanced mixing. Reactor temperature increased toward midday as solar irradiance increased and then gradually decreased during the afternoon. The maximum temperatures were approximately 298.06 K at 09:00, 298.12 K at 12:00, and 298.03 K at 16:00. Axial analysis showed gradual heat accumulation along the reactor, resulting in higher outlet temperatures than inlet temperatures.

16. On the Bi-parametric Stability of High Accuracy ADI Method of $O(2,4,4)$ for 2D Model Transmission Line Equation on an Unequal Mesh

R.K. Mohanty

Department of Mathematics, South Asian University, New Delhi, India

Abstract: An alternating direction implicit (ADI) method of order of accuracy $(2,4,4)$ is suggested for 2D Transmission Line Equation on an unequal mesh. The method is stable for all choices of step-lengths in x -, y - and t -directions. First, we use two-parameter mesh ratios and talk about a new three-level compact implicit method of $O(2,4,4)$ for the solution of 2D second order nonlinear hyperbolic PDEs. The numerical approximation at first time level is briefly discussed. Various benchmark problems are solved to confirm the usefulness of the proposed numerical algorithms.

17. Modeling the Spread of Neolithic Agriculture through Go-or-Grow Dynamics [P-13]

M. Humayun Kabir

Jahangirnagar University, Bangladesh

Abstract: Archaeological evidence suggests that the spread of Neolithic agriculture was a gradual transition from nomadic foraging to sedentary farming, with an approximately constant rate of expansion across different regions [1]. While numerous mathematical models have been proposed to explain this process, most predict expansion speeds that are considerably faster than those inferred from archaeological observations [1]. To address this discrepancy, we develop a three-component reaction-diffusion/ODE model that distinguishes between two classes of farmers, migratory and sedentary, and incorporates transitions between these behavioral states [2]. We further extend the model by incorporating the influence of farming technology on the dynamics of agricultural expansion, aiming to elucidate its role in determining the rate of spread. In this talk, I will present an analysis of one-dimensional traveling wave solutions and characterize the minimum wave speed under the assumption of radially symmetric expansion. Our results demonstrate that the minimum traveling wave speed provides a plausible explanation for the observed rate of Neolithic agricultural spread. Numerical simulations further reveal that advances in farming technology can significantly reduce the minimum wave speed, leading to slower and more realistic expansion dynamics consistent with archaeological evidence [3]. Finally, I will discuss the planar stability of the minimum traveling waves and investigate the emergence of two-dimensional traveling-wave solutions, highlighting how technological development influences both the stability and the spatial propagation patterns of agricultural expansion [4].

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18. Mathematical Modelling of Tumor-Obstructed Non-Newtonian Biofluid Transport

Dhamendra Tripathi

National Institute of Technology, Uttarakhand, India

Abstract: Tumor growth progressively narrows physiological channels and significantly alters biofluid transport, particularly when the fluid exhibits non-Newtonian behavior. This talk presents a mathematical model for tumor-obstructed transport of a micropolar biofluid in a peristaltically driven channel, where the tumor is represented by a time-dependent, exponentially growing geometric constriction. Under long-wavelength and low-Reynolds-number approximations, analytical solutions are derived to examine velocity distribution, pressure gradient, microrotation, wall shear stress, and particle transport. The analysis demonstrates that tumor progression intensifies channel constriction, suppresses

forward flow, induces localized reversal, and elevates pathological stresses near the obstruction. Micropolar parameters significantly influence transport characteristics, highlighting the importance of microstructural effects in physiological fluids. The study provides theoretical insight into cancer-induced transport abnormalities and offers potential applications in predicting obstruction severity, optimizing drug delivery, and improving healthcare strategies.

19. Asymmetric Steady States and Population Recovery in a Delayed Patchy Model

Bapan Ghosh

IIT Indore, India

Abstract: Populations occupy in fragmented habitats and disperse from one cite to another. Populations spent some time to travel from one location to another which is termed as travel time (dispersal delay). We develop a predator-prey model where local patches (nodes) are connected via dispersal and time delay. We explore how delay and dispersal can shape population dynamics. The isolated patch can exhibits at most two coexisting equilibria. However, dispersal may generate multiple asymmetric stable equilibria inducing bistability and tri-stability among coexisting states. A key result is the existence of a stable coexisting asymmetric pair even when the corresponding within-patch system admits no stable coexistence steady state. This reflects the positive role of dispersal in recovering populations from extinction. Various stability classifications are presented to highlight the effect of increasing dispersal delay on the combined stability behavior of symmetric and asymmetric equilibrium pairs. The mathematical tools, computational techniques, and ecological principles derived from the model might induced new knowledge in this domain.

20. Multi-region evaluation of a hybrid ARIMAX–ARNN model for dengue incidence forecasting

Dr. Sourav Rana

IEST, Shibpur

Abstract: Hybrid modelling approaches for dengue incidence forecasting have gained significant attention due to the increased availability of epidemiological and climatic data. Accurate forecasting of dengue incidence is essential for early intervention planning, resource allocation, and outbreak preparedness. Recent studies have explored combining statistical time series methods with machine learning techniques for predicting dengue cases, achieving varying degrees of success. In this study, we introduce an extended hybrid ARIMA-ARNN (Autoregressive Integrated Moving Average - Autoregressive Neural Network) model aimed at dengue incidence forecasting using rainfall as an exogenous variable. The primary goal is to propose this hybrid model and evaluate its forecasting performance. The asymptotic stationarity of the proposed model is discussed based on existing theoretical frameworks. Weekly dengue incidence data from Colombia, Iquitos, San Juan and Singapore are analyzed to assess forecasting performance. The performance of the proposed model is compared with benchmark forecasting methods using standard accuracy metrics under a 4-week-ahead forecasting horizon. The results demonstrate that the proposed model outperforms the ARIMAX model in three out of four tested locations, with improvements observed across

MAE, RMSE and MAPE. Both theoretical insights and empirical findings indicate the usefulness of the proposed model for dengue incidence forecasting.

21. Decision Analysis of Insurance Providers Using Double-Fold MEREC Weighting and MARCOS under Neutrosophic Information **[I-09]**

Biplab Sinha Mahapatra

Haldia Institute of Technology, India

Abstract: Existing MCDM techniques often treat sub-criteria within a criterion as independent criteria when determining criterion weights, thereby overlooking their individual characteristics. To address this limitation, this study develops a double-fold weight determination concept based on the Removal Effects of Criteria (MEREC) method. In the first fold, MEREC independently determines the weights of sub-criteria within each criterion, reflecting their individual significance. In the second fold, criterion weights are derived from the accumulated decision matrix. To handle uncertainty in multi-criteria decision-making, a neutrosophic set-based Measurement Alternatives and Ranking according to the COMpromise Solution (MARCOS) approach is developed for alternative ranking. The proposed framework is applied to assess the sustainability of health insurance providers (HIPs) from an Indian perspective. Its performance is compared with single MEREC integrated with neutrosophic MARCOS to examine differences in ranking outcomes. Various decision-making scenarios are further considered by varying criterion weights. The results show that double-fold MEREC provides an approximately 14% wider range of performance measures than single MEREC, supporting more stable and credible rankings for decision makers.

❖ Mathematical Analysis (Pure Mathematics):

1. Introduction to random matrices and non-commutative probability **[P-02]**

Arup Bose

Stat-Math unit, Indian Statistical Institute, Kolkata, India

Abstract: High dimensional random matrices and non-commutative probability are inextricably linked via the results of Dan Voiculescu on the asymptotic freeness of independent Wigner matrices. We shall start with the basic results on classical moments and cumulants using the Mobius function on the class of all partitions. We will then introduce free cumulants via non-crossing partitions, leading to the idea of freeness in non-commutative

probability spaces. Asymptotic freeness results for some random matrices, including Wigner and sample covariance matrices will be explained.

2. On the Betti numbers of a homogeneous space of a Lie group

[P-03]

Pralay Chatterjee

IMSc, Chennai, India

Abstract: We will describe our work in collaboration with I. Biswas and C. Maity. We will begin the talk by indicating how the exactness criteria of the Kostant-Kirillov symplectic form on homogeneous spaces of a semisimple Lie group led to the question of computing second cohomology of such spaces. We will then give computable descriptions of the first four Betti numbers of a homogeneous space of a Lie group. We will also indicate how an interesting topological invariant is obtained using certain Lie theoretic data associated to a homogeneous space. We will apply the results when the homogeneous spaces are nilpotent orbits in a semisimple Lie group.

3. Approximation and interpolation

[P-11]

Jaydeb Sarkar

Indian Statistical Institute, Kolkata, India

Abstract: The first part of this talk will focus on several classical results in approximation and interpolation theory that have had a significant impact on analytic function theory and functional analysis. The second part will address some of the known challenges in extending these results to several complex variables. The final part will present recent findings that help address these difficulties to some extent.

4. Deformation space of Riemann surfaces: Analysis and Geometry.

[P-09]

Hideki Miyachi

School of Mathematics and Physics, College of Science and Engineering, Kanazawa University, Kakuma-machi, Kanazawa, Ishikawa, Japan

Abstract: The moduli space of Riemann surfaces parametrizes Riemann surfaces up to conformal equivalence and is closely connected to many areas of mathematics. It is obtained as the quotient of Teichmüller space by the action of the mapping class group. In this talk, I will present my recent work on the analytic and geometric properties of Teichmüller space for

closed Riemann surfaces. I will first review some basic results on Teichmüller space and then discuss its complex-analytic properties.

5. Twisted Patterson–Sullivan Densities on Nilpotent Covers for Anosov Subgroups. [P-08]

Krishnendu Gongopadhyay

IISER Mohali, India

Abstract: Counting the orbit of a group in the hyperbolic plane produces a natural measure on the boundary at infinity. We begin with this classical picture, then ask what changes when orbit points are weighted or the space is replaced by an infinite cover. For higher-rank Anosov groups, we discuss the existence, uniqueness and ergodicity of the resulting measures, their classification on certain covers, and when a cover preserves the rate of orbit growth. No background in hyperbolic geometry is assumed. The talk is based on a joint work with Neelanjan Mondal.

6. An improvement of the Blanco-Koldobsky-Turnšek characterization of isometries

Debmalya Sain

Indian Institute of Information Technology Raichur, India

Abstract: The classical Blanco-Koldobsky-Turnšek Theorem characterizes the isometries on a Banach space as norm one linear maps that preserve Birkhoff-James orthogonality. Very recently, we have embarked on a program towards a better understanding of this seminal result. In this talk, I shall consider a local version of the preservation of orthogonality by linear operators between Banach spaces and discuss its consequences. As an application of the local approach, we will improve the nontrivial direction of the said characterization in its most general setting.

Young Scientist Symposium

1. Equi-Baire One Property for $SL(3, \mathbb{H})$ actions on the Quaternionic Projective Plane [S-01]

Sandipan Dutta

Mizoram University, Aizawl, India

Abstract: We study projective transformations induced by matrices in $SL(3, \mathbb{H})$ acting on the quaternionic projective plane. For maps admitting an attracting fixed point and a repelling hyperplane, we prove that the family of iterates satisfies the orbitally Equi-Baire one property and introduce a function describing orbit convergence. We also analyze one-parameter subgroups and obtain a dichotomy: relatively compact subgroups generate regular behaviour,

while non-compact subgroups collapse open sets onto projective subspaces, destroying the Equi-Baire one property. These results connect algebraic properties of quaternionic linear groups with dynamical behaviour and limit structures in quaternionic projective geometry.

2. Distance to the Nearest Structured Polynomial Matrix with a Prescribed Eigenvalue Geometric Multiplicity

[S-02]

Tanay Saha

Mizoram University, Aizawl, India

Abstract: We study the problem of finding the nearest structured regular polynomial matrix to a given structured polynomial matrix such that the perturbed matrix possesses an eigenvalue with prescribed geometric multiplicity. We first consider linearly structured unimodular polynomial matrices without imposing explicit constraints on the eigenstructure of the perturbed non-unimodular matrix. We then extend the framework to general linearly structured polynomial matrices by incorporating constraints on the eigenstructure. In particular, we focus on computing the nearest structured regular polynomial matrix having an eigenvalue whose geometric multiplicity satisfies a prescribed lower bound. The eigenvalue is initially allowed to be any complex number, followed by a more specialized treatment in which the eigenvalue is required to be real. The proposed approach formulates these nearest-matrix problems as constrained optimization problems, enabling their numerical solution through established optimization techniques. Several numerical experiments are presented to illustrate the effectiveness, robustness, and practical applicability of the proposed methodology.

3. Pulse-Induced Nonautonomous Dynamical Systems in Epidemiology: Modelling, Thresholds, and Disease Dynamics

[S-03]

Amit Kumar Roy

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Many infectious diseases exhibit pronounced temporal variability arising from seasonality, climatic fluctuations, behavioural changes, population movement, and episodic public-health interventions. Such mechanisms naturally lead to nonautonomous epidemiological systems in which transmission or intervention-related parameters vary explicitly with time. This work focuses on pulse-induced nonautonomous epidemic models and their role in describing recurrent, seasonal, and outbreak-driven disease dynamics. In particular, two forms of temporal forcing: periodic harmonic (cosine) functions, Gaussian-type pulses, have been considered. This work investigates how the amplitude, timing, frequency, and width of external pulses influence epidemic peaks, persistence, extinction thresholds, and long-term dynamical behaviour. Analytical concepts involving disease-free and endemic states, threshold quantities, stability, and periodic solutions will be presented together with numerical simulation and data-driven parameter estimation. The work also illustrates how pulse-based forcing can provide a flexible alternative to conventional sinusoidal seasonality when transmission is concentrated within relatively short epidemic windows. Finally, the potential of such models for disease forecasting, evaluation of

time-dependent interventions, and characterization of region-specific epidemic patterns will be discussed, highlighting emerging directions in nonautonomous mathematical epidemiology.

4.

5. Role of Stem Cells in Skin Recovery: A Mathematical Approach to Understanding Psoriasis and Its Control [S-05]

Subhankar Kushary

Department of Mathematics, Brainware University, Kolkata, West Bengal, India

Abstract: Psoriasis is a chronic, immune-mediated inflammatory skin disease caused by complex interactions between immune cells, cytokines, and keratinocytes. These interactions lead to abnormal growth of epidermal cells and inflammation, resulting in red, scaly patches on the skin. The proposed research aims to develop a mathematical framework to understand the progression behavior of psoriasis by incorporating stem cell dynamics, immune-cell interactions, keratinocyte proliferation, and cytokine-mediated signaling pathways. Special attention will be given to the immunomodulatory role of mesenchymal stem cells (MSCs) and their potential effects on pro-inflammatory and anti-inflammatory cytokine networks. The research will investigate how stem cell-mediated feedback mechanisms influence disease progression, system stability, and the occurrence of psoriasis flare-ups. Sensitivity analysis of the key model parameters, along with numerical simulations, will be used to identify important biological factors associated with disease persistence and inflammatory regulation. Furthermore, impulsive control strategies will be explored to study the potential effects of stem cell-based interventions and compare them with conventional cytokine-targeted therapies. The proposed framework aims to provide a mathematical and computational basis for understanding stem cell-mediated immunoregulation and evaluating potential therapeutic strategies for psoriasis.

6. Recent Advancements in the Theory of Characterized Subgroups [S-06]

Ayan Ghosh

NIT Durgapur, India

Abstract: The theory of characterized subgroups of the circle group has seen significant recent progress, particularly in the context of explicit constructions. In recent joint work, the introduction of arithmetic-type sequences has enabled a more refined structural analysis. It is shown that, within this framework, a characterized subgroup is countable if and only if it is torsion. Furthermore, every infinite torsion subgroup of the circle group can be characterized by an arithmetic-type sequence with bounded ratio. In subsequent work, we provide a characterization of those arithmetic-type sequences that yield countable characterized subgroups, thereby identifying a natural subclass that captures all infinite torsion subgroups. Our results also demonstrate that the classical dichotomy in H. G. Eggleston's theorem for arithmetic sequences does not extend in general to arithmetic-type sequences; however, a stronger dichotomic phenomenon emerges within this broader setting.

7. Rate-induced phase tipping in birhythmic oscillators [S-07]

Ravi Kumar K

Indian Institute of Technology Ropar, India

Abstract: This presentation discusses rate-induced phase-tipping (RP-tipping) between two stable limit cycles of a birhythmic oscillator. Such an oscillator RP-tips when a time variation of an input parameter preserves the bistability of the limit cycles but induces transitions from one stable limit cycle to the other, causing abrupt changes in the amplitude and frequency of the oscillations. Crucially, these transitions occur when: the rate of change of the input is in a certain interval bounded by critical rate(s), and the system is in certain phases of the limit cycle. An illustrative example of the birhythmic van der Pol oscillator model is used along with monotone time-varying external forcing to demonstrate this RP-tipping.

8. On the preservation of parallel pairs and triangle-equality attainment pairs, with applications to isometries [S-08]

Kalidas Mandal

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: We investigate the structure and preservation of parallel pairs and triangle equality attainment (TEA) pairs in normed linear spaces. We begin by providing functional characterizations of these pairs in normed linear spaces and provide a characterization of finite-dimensional Banach spaces with numerical index one. Next, we show that in finite-dimensional Banach spaces, a bijective bounded linear operator preserves TEA pairs if and only if it preserves parallel pairs, and this equivalence remains valid in finite-dimensional polyhedral spaces for operators with rank greater than one. We further explore some geometric consequences related to this preservation and provide a characterization of isometries on certain polyhedral Banach spaces as norm-one bijective operators that preserve the norm at extreme points and also preserve the parallel or TEA pairs.

9. On singquandles arising from Core Quandles and their shadows [S-09]

Dr Abhishek Mukherjee

Kalna College, India

Abstract: A quandle R is meant to be a right distributive quasi groupoid in which every element is idempotent. Based on these algebraic systems, a non-associative algebra, namely Oriented Singquandle, will be introduced in the meantime, motivated from the oriented singular knots having regular and singular crossings. Notably, people have developed ample amount of examples of oriented singquandles that are

based on the conjugation quandle $\text{Conj}(G)$. In this talk, we focus on manufacturing some new examples of oriented singquandles out of the core quandle $\text{Core}(G)$. Also we try to highlight algebraic interventions of those singquandles in the theory of rings and modules. Lastly, we will try to develop an idea to produce singquandle shadow in a general setting. This talk is based on our recent work with Prof. Sujit Kr. Sardar and Mr. Siddhartha Banerjee.

10. From Disease Dynamics to Sustainability: Mathematical Modelling of Complex Biological, Computational and Environmental Systems [S-10]

Dr. Salil Ghosh

Abstract: This talk presents selected mathematical, computational, and data-driven approaches for addressing contemporary problems in disease dynamics, therapeutic optimization, epidemiology, ecological management, renewable energy, and environmental sustainability. Particular emphasis is placed on nonlinear dynamical systems, stochastic and fractional-order modelling, periodic systems, optimal control, sensitivity analysis, parameter estimation, and real-world data validation. Applications include within-host leprosy dynamics, combination therapies for cervical cancer, seasonal transmission of human metapneumovirus, zoonotic cutaneous leishmaniasis, and sustainable pest management through sterile insect techniques. The study also extends to ultrasound-assisted biodiesel production and deep decarbonization of the building sector through integrated energy-system modelling. Across these diverse applications, mathematical models are used to identify critical mechanisms and thresholds, assess the influence of uncertainty and heterogeneity, evaluate intervention strategies, and support cost-effective decision making. The results highlight the broad potential of integrated mathematical and computational frameworks for understanding complex biological, ecological, energy, and sustainability-related systems and for translating theoretical insights into practical scientific guidance.

11. FEM utilizing the HHT- α Method for Nonlinear Dynamic Buckling of Imperfection-Sensitive Columns [S-10]

Rajib Karmakar

Abstract: This study presents a mathematical and computational framework for analyzing the nonlinear dynamic bifurcation behavior of continuous cantilevered structures under transient loading, applied to reinforced cement concrete column. Using Hamilton's principle, a system of coupled, nonlinear partial differential equations extending the classical Euler buckling formulation is derived. This model incorporates perturbation parameters for initial geometric imperfections, with nonlinear constitutive operators for stiffness degradation and structural damping. To solve the governing boundary value problem, finite element spatial discretization is employed. Temporal evolution is integrated via the Hilber-Hughes-Taylor (HHT- α) implicit algorithm, chosen to preserve numerical stability and ensure convergence during highly nonlinear transient regimes. Numerical analysis reveals profound sensitivity of dynamic instability thresholds to initial perturbations. Introducing a geometric imperfection amplitude of δ_0 reduces the critical dynamic bifurcation load relative to the unperturbed static state. Parametric studies demonstrate that increasing the slenderness ratio (λ) drops the

critical load causing amplified lateral displacements. Transient forcing near resonant frequencies contracts the stability envelope by nearly 22%. Simulating concrete cracking via a degraded stiffness matrix captures an additional 10-14% reduction in stability margins. Finally, analysis shows circular domains expand the dynamic stability basin by 24% over equivalent square domains.

Contributory Speakers

❖ Analysis-Algebra-Geometry-Combinatorics-Graph Theory:

1. q-Berezin sectorial operators and their applications to q-Berezin number inequalities. [C-10]

Saikat Mahapatra

Department of Mathematics, Jadavpur University, Kolkata

Abstract: Here, we introduce a new class of operators, called q-Berezin sectorial operators, which extends the class of q-sectorial operators. Through an example on the Hardy-Hilbert space, we show that a q-Berezin sectorial operator need not be q-sectorial. Furthermore, we establish several new inequalities for the q-Berezin number associated with this class of operators.

2. On the Thickness of Line Graphs of Intersection Graphs of Ideals [C-03]

Gargi Kuila

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Abstract: Let R be a commutative ring with unity and $I^*(R)$ denote the set of all nontrivial ideals of R . The intersection graph of ideals of R , denoted by $G(R)$, is the simple graph whose vertex set is $I^*(R)$, where two distinct ideals I and J are adjacent if and only if $I \cap J \neq (0)$. In this work, we focus on graph theoretic aspects of line graphs arising from intersection graphs of ideals of rings. We determine bounds for the thickness and outer thickness of $L(K_n)$ and further investigate the thickness and outer thickness of $L(K_{(2,n)})$. Finally, we characterize the values of n for which the line graph $L(G(Z_n))$ is Eulerian or Hamiltonian.

3. On Graded m -Nil Clean Rings

[C-06]

Saikat Das

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Abstract: In this presentation, we introduce the concept of graded m -nil clean ring to extend the existing notion of graded nil-clean ring introduced in [1]. We explore fundamental properties of these rings, emphasizing the interplay between the identity component and the graded structure. We investigate certain conditions under which graded group rings, graded matrix rings and graded amalgamated algebras inherit the m -nil clean property from their components. Specifically, we establish sufficient conditions for graded group rings and graded matrix rings over commutative m -nil clean rings to be graded m -nil clean rings.

Reference: [1] Ilić-Georgijević, E. & Sahinkaya, S. (2018) : On graded nil clean rings, Communications in Algebra, 46(9), 4079-4089

4. A study of planarity in bipartite graphs defined by group elements and subgroups

[C-14]

Munna Pramanik

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Abstract: To each non-trivial group G , we associate a simple bipartite graph $\Gamma(G)$ constructed from the elements and the non-identity subgroups of G . The bipartition of $\Gamma(G)$ consists of the set $A=G$ and the set B of all non-trivial subgroups of G . The incidence relation is defined by declaring an element $a \in A$ and a subgroup $H \in B$ to be adjacent whenever the square a^2 belongs to H . The primary objective of this work is to investigate the structural and embedding properties of $\Gamma(G)$ in relation to the algebraic structure of G . For finite groups, we give a characterization of those groups for which the associated graph is planar. We also obtain complete descriptions of the finite groups whose corresponding graphs are outerplanar, maximal planar, maximal outerplanar, stars, trees, or claw graphs. Finally, we extend the investigation beyond the finite setting by studying the planarity of $\Gamma(G)$ for infinite groups and deriving conditions governing this property.

5. On Factorization of Polynomial matrices

[C-38]

Anjali Beniwal

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Abstract: In this work, we reduce the computation of a Greatest Common Divisor (GCD) of polynomial matrices to the factorization of constant, structured generalized Sylvester matrices. Specifically, we show that the null space of a suitably constructed generalized Sylvester matrix characterizes the GCD of a given collection of polynomial matrices. Based on this characterization, the coefficients of the GCD can be recovered directly by applying standard numerical linear algebra tools, such as the QR factorization, to the structured matrix, without explicitly computing a full factorization of the polynomial matrices.

6. Generating Sequence-Based SLRA for Toeplitz and Hankel Matrices [C-40]

Pranab Modak

Research Scholar, India

Abstract: Structured Low-Rank Approximation (SLRA) is a fundamental tool in numerical linear algebra with widespread applications in signal processing, system identification, and data analysis. While numerous frameworks exist in the literature to tackle this problem, finding efficient and numerically stable solutions for specific matrix structures remains challenging. In this work, we present a novel approach to SLRA for Hankel and Toeplitz structured matrices based on generating sequences. By leveraging the algebraic properties of generating sequences, our method simplifies the underlying optimization problem, offering an alternative computational strategy with improved accuracy and efficiency. We discuss and demonstrate its effectiveness through numerical experiments, comparing its performance against established state-of-the-art methods.

7. On the Distance to Non-Primeness for Affine Structured Polynomial Matrices: Optimization Approach [C-41]

Parna Saha

Department of Mathematics and Computer Science, Mizoram University

Abstract: Computing the minimal distance from a structured left-prime polynomial matrix to the nearest non-prime matrix is a fundamental problem in robust control theory and signal processing. Existing Structured Low-Rank Approximation (SLRA) approaches fail to guarantee structure preservation in some cases and break down when handling rank-deficient leading coefficient matrices. To address these limitations, this paper establishes a unified theoretical framework for polynomial matrices subject to general linear and affine structures. We derive formal existence conditions for perturbations that induce non-primeness while strictly preserving arbitrary linear and affine coefficient constraints, even when the leading coefficient matrix is rank-deficient. Based on these theoretical insights, the problem is

formulated as a constrained nonlinear optimization program that minimizes the Frobenius norm of the perturbation. Finally, extensive numerical experiments across various linear and affine structures, alongside control-theoretic applications such as determining the structured radius of controllability, validate the theoretical findings and demonstrate the efficacy of the proposed optimization approach.

❖ Mathematical Modelling & Mathematical Biology:

8. Patient-Data-Driven HIV Dynamics with Macrophage-Mediated Reinfection Using Noise-Induced Physics-Informed Neural Networks [C-02]

Purnendu Sardar

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Human immunodeficiency virus (HIV) continues to pose a significant public-health challenge due to the complex interplay among immune-cell dynamics, viral replication, and persistent cellular reservoirs. In this work, we develop and analyze a nonlinear HIV infection model incorporating healthy and infected CD4⁺ T cells, free virus, and an explicit macrophage reservoir, together with density-dependent mortality of CD4⁺ T cells and macrophage-mediated reinfection. The mathematical properties of the model are established, followed by a detailed investigation of its equilibrium states, local stability, sensitivity, and bifurcation behavior. The analysis identifies transcritical, saddle-node, and Hopf bifurcations, indicating transitions among viral clearance, persistent infection, and oscillatory regimes. Sensitivity analysis highlights infection and viral production rates as key determinants of HIV progression, while viral clearance plays an important role in infection control. To account for environmental and biological variability, stochastic perturbations are incorporated into the model. The resulting simulations demonstrate that weak fluctuations may preserve viral persistence, whereas sufficiently strong stochastic effects can substantially reduce infected CD4⁺ T cells, viral load, and macrophage-associated viral reservoirs. Furthermore, a physics-informed neural network (PINN) framework is developed to reconstruct the underlying disease dynamics and estimate unknown model parameters from noisy observations. The proposed framework exhibits robust performance in the presence of observational uncertainty. Finally, real patient data are integrated to assess the model's predictive capability, showing good agreement with observed CD4⁺ T-cell counts and viral-load trajectories and indicating considerable heterogeneity in macrophage reservoir activity across patients. The combined mathematical, stochastic, and physics-informed learning framework provides a promising approach for patient-specific characterization of HIV infection dynamics and may support improved understanding of viral persistence and treatment strategies.

9. Threshold Dynamics and Complex Behavior in a Nonlinear Impulsive Model of Oncolytic Virotherapy [C-16]

Imran Ali

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Oncolytic virotherapy offers a targeted cancer treatment by selectively infecting and destroying malignant cells while preserving healthy tissue. In this study, we develop a nonlinear impulsive mathematical model to analyze interactions between uninfected and infected cancer cells (UCC and ICC) under periodic oncolytic virotherapy. The model incorporates logistic tumor growth, nonlinear cell-to-cell transmission, and impulsive conversion of susceptible cells, providing a biologically realistic yet analytically tractable framework. We derive an explicit threshold for the dosing interval that determines treatment outcome: below this threshold, the cancer-free periodic solution is globally asymptotically stable, ensuring tumor eradication, whereas above it, the system exhibits persistence. Numerical simulations and bifurcation analysis reveal a rich range of dynamics, including periodic, multi-periodic, and chaotic oscillations driven by treatment frequency and infection efficiency. These results highlight that therapeutic success depends on the interplay between dosing schedule and viral effectiveness. The study provides new insights into tumor-virus dynamics and offers a theoretical basis for optimizing oncolytic virotherapy protocols.

10. AMP Feedback Loop in Psoriasis Dynamics: Bifurcation and Stability Analysis of a Th17-Th2-Keratinocyte-Cytokine Model [C-18]

Naaz Khan

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: The paper on AMP Feedback Loop in Psoriasis Dynamics: Bifurcation and Stability Analysis of a Th17-Th2-Keratinocyte-Cytokine Model presents a four-dimensional mathematical model that couples Th17 cells, Th2 cells, keratinocytes, and pro-inflammatory cytokines. Th17 activation has been represented through a Hill function to capture its receptor-driven, saturating response to cytokine and AMP levels. Homeostatic and high inflammatory states in psoriasis have been established, and explicit threshold conditions on the AMP level have been derived. Local stability has been examined using the Routh-Hurwitz criteria, and this has produced AMP thresholds that separate homeostatic, mild and high inflammatory regimes. Hill coefficients $n = 1$ and $n = 2$ have been compared, and this has shown that sharper Th17 activation producing a distinct bistable window and hysteresis between mild and severe disease states for higher values of Hill function comparatively to its lower value. The sensitivity analysis has shown that different parameters dominate under low and high AMP conditions, with Th2-regulatory and keratinocyte proliferation parameters governing the system at low AMP levels, while AMP-dependent Th17 activation parameters become more influential at high AMP levels. These results have identified AMP level as a threshold-driven control parameter and have provided a

mathematical basis for distinguishing monostable and bistable inflammatory regimes in psoriasis progression.

11. A Unified Framework for Lyapunov Function Construction in Generalized Stochastic Population-Growth Models [C-19]

Parthib Sengupta

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Environmental variability, demographic fluctuations, and resource uncertainty can strongly affect the stability and persistence of biological populations. Lyapunov's second method, together with the infinitesimal Markov generator, provides a powerful framework for analysing stochastic stability without requiring explicit solutions of stochastic differential equations (SDEs). However, constructing suitable Lyapunov functions is often model-specific. In this study, we develop a unified framework for constructing Lyapunov functions for a generalized class of nonlinear stochastic population-growth models. By exploiting the underlying biological mechanisms of density regulation, cooperation, and intra-specific competition, we introduce a systematic transformation that reduces the nonlinear dynamics to a form with a natural multiplicative structure, enabling a corresponding Lyapunov construction. Using Itô's formula and the associated Markov generator, we derive a sufficient condition on environmental noise intensity for stability in probability. The framework encompasses several classical models, including stochastic logistic and theta-logistic growth, and provides a systematic alternative to model-specific Lyapunov function selection for analysing stochastic population dynamics.

12. Adaptive Contact Behavior under Awareness Campaigns: A Case Study of an Epi-Economic Model of Leprosy [C-20]

Tarun Mondal

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Neglected tropical diseases (NTDs) remain difficult to diagnose due to their long incubation periods. Furthermore, frequent asymptomatic transmission associated with NTDs makes intervention strategies difficult to implement effectively. NTDs are also associated with visible deformities and disabilities, which contribute to persistent societal stigma and significantly reduce the quality of life of infected individuals. Furthermore, increasing digitalization has facilitated the rapid spread of misinformation, amplifying public fear and misconceptions surrounding these diseases.

Considering these factors, we develop a four-dimensional mathematical model within a forward-looking socio-economic behavioral framework. In this setting, individuals adjust their contact preferences based on perceived incentives from social interaction while accounting for the potential risk of infection. Hence, contact decisions arise from the trade-off between social or economic benefits and the expected utility costs associated with risky interactions. This epi-economic framework allows behavioral adaptation to influence

the epidemic trajectory over time.

Numerical simulations, considering disease leprosy as a representative case, indicate that adaptive behavioral responses can substantially reduce the prevalence of symptomatic infections. However, the persistence of asymptomatic carriers continues to sustain disease transmission, suggesting that asymptomatic spread remains a major challenge for disease eradication. Our findings further indicate that periodic awareness campaigns can improve societal perceptions of infection and contribute to enhancing the quality of life of affected individuals.

13. Harvesting Reshapes Resource Enrichment and Spatial Self-Organization in a Resource-Mediated Intraguild Predation System [C-21]

Sourav Maity

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Understanding how harvesting interacts with resource availability and spatial processes to regulate ecosystem dynamics is a fundamental problem in theoretical ecology and fisheries management. We develop and analyze a reaction--diffusion resource-mediated intraguild predation(IGP) model for zooplankton, prey fish, and predator fish populations that incorporates alternative resource support and nonlinear saturated harvesting of both fish species. The model is used to investigate how harvesting, basal resource enrichment, and spatial diffusion jointly influence species coexistence, ecosystem stability, and spatial organization. Temporal analysis reveals that increasing prey harvesting drives the system from stable coexistence to oscillatory and chaotic dynamics through Hopf and period-doubling bifurcations, whereas predator harvesting induces transcritical bifurcations between alternative equilibrium states by modifying top-down regulation. Increasing zooplankton productivity destabilizes the coexistence equilibrium through a Hopf bifurcation, providing a manifestation of the paradox of enrichment. Incorporating spatial diffusion reveals stable, Turing, Hopf, and combined Hopf-Turing regimes that generate a rich variety of self-organized spatial patterns, including spots, stripes, labyrinthine structures, and complex spatiotemporal dynamics. We show that harvesting not only regulates population abundance but also modifies the ecological conditions under which resource enrichment and spatial diffusion generate temporal oscillations and spatial heterogeneity. Consequently, harvesting, dispersal, and basal resource productivity jointly determine ecosystem resilience, species persistence, and pattern formation. These findings provide new mechanistic insights into the coupled ecological and anthropogenic processes governing exploited aquatic ecosystems and highlight the importance of integrating harvesting and spatial dynamics into ecosystem-based management.

14. A Fractional-Order Commensurate and Incommensurate Study on Predator-Prey System with Allee in Predator and Fear Effect [C-23]

Arnabi Saha

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: As days goes, the demand of vehicles increases, while availability of raw materials decreases. At a result, raw materials and fuel. In addition, fuel-based vehicles emit carbon dioxide continuously. As a result, environmental sustainability decreases. Addressing it, industry tries to find alternative ways so that it conserves raw material, which are used for build vehicles, saves fuel, maintains sustainability. Motivated from, this paper develops a stochastic-demand closed-loop supply chain model in which eco-friendly products are manufactured using recovered materials obtained from end-of-life products. The proposed model incorporates two product return strategies, Trade-Old-for-Cash and Trade-Old-for-New, motivate customers to return old products. In addition, shortage and refund policies are considered to investigate their effects on supply chain performance. The objective is to increase the industry's profit while ensuring resource conservation, promoting green innovation in production, and maintaining environmental sustainability. The formulated maximization problem is optimized, and several special cases are analyzed to compare economic benefits. A numerical study is conducted to illustrate the applicability of the proposed model, analyze several cases, and validate the theoretical findings. Furthermore, sensitivity analysis is performed to examine the impact of key parameters on the optimal decisions. The analysis reveals that applicable of trade-old-for-cash, trade-old-for-new has a positive impact, and the refund policy negatively affects the industry's profitability under the considered settings. The proposed model useful insights for industries into the implementation of trade-old-for-cash, trade-old-for-new programs, shortage management, and sustainable closed-loop supply chain operations

15. Analyzing Control Strategies in DC/DC Buck Converters: A Mathematical and Simulink-Based Approach [C-28]

Oindrella Dey

Dept. of EEE, Premier University, Bangladesh

Abstract: The DC/DC buck converter is a fundamental power electronics topology for voltage step-down operations. While Proportional-Integral-Derivative (PID) controllers are conventionally used for output regulation, they frequently struggle to manage nonlinear switching dynamics and parameter variations. To address these limitations, this study utilizes small-signal modeling to approximate the nonlinear converter as a linear system, establishing a benchmark between conventional PID and advanced methodologies specifically fuzzy logic and adaptive control algorithms. These mathematical models are rigorously validated within MATLAB/Simulink to evaluate dynamic transient behavior and frequency response. Simulation results demonstrate that while conventional PID control achieves baseline steady-state accuracy, it exhibits significant voltage overshoot during a 12V to 7.9V step-down and prolonged settling times under sudden load disturbances. In contrast, the integration of fuzzy logic effectively mitigates these nonlinearities, yielding an reduction in voltage overshoot (around 85%) and 40% improvement in settling time. Furthermore, adaptive control mechanisms enhance reliable system stability and preserve energy efficiency, even amid hardware parameter fluctuations such as shifting inductance or capacitance. Ultimately, integrating foundational mathematical modeling with comprehensive

simulation provides a highly effective framework for the analysis and design of next generation power converters.

16. Top-down regulation and catastrophic transition in an intraguild predation model driven by IG-prey's anti-predator nature [C-29]

Debabrata Mandal

Indian Institute Of Technology, Roorkee, India

Abstract: Intraguild predation (IGP) combines predation and competition. The significance of IGP-models in many natural groups is becoming increasingly clear. In these IGP-systems, we sometimes witness a notable ecological phenomenon called top-down regulation, in which predatory activity at a higher trophic level influences the community structure of lower trophic levels. This talk focuses on how the anti-predator behavior of dangerous IG-prey regulates the top-down effect in an ecosystem. To accomplish this, we developed a three-layered IGP-model incorporating the Holling type-I and type-IV functional responses for interactions of shared-prey with their predators and between IG-prey and IG-predators, respectively, along with the anti-predator behavior of IG-prey through direct killing of primarily juvenile predators. This study provides adequate requirements for all equilibria to exist and be locally stable for the proposed IGP-model. Sufficient conditions for species persistence and extinction are discussed analytically. Depending on the initial population size, the IGP-model exhibits hysteresis cycle and undergoes a catastrophic shift to extinction of the IG-predators by altering the strength of the IG-prey's group defense. Hysteresis cycles occur when the system follows multiple transition paths during forward and reverse changes in a parameter, resulting in different stable states depending on the initial conditions within a parameter range. On the other hand, a catastrophic shift is an abrupt and irreversible change in a system's state caused by a slight change in a control parameter, such as a saddle-node bifurcation, causing species extinction. In a nutshell, our IGP-model fails to show top-down regulation in the presence of the dangerous IG-prey's anti-predator nature, but in its absence, it does.

17. Periodic Structures, Period-Bubbling, and Multistability in a Fractional-Order Ecological System with Allee Effect [C-30]

Prajwal Gupta

Banaras Hindu University, Varanasi, India

Abstract: Nonlinear ecological systems can display complex behaviors, including organized periodic structures, abrupt transitions between attractors, period-bubbling, and multistability. Although these phenomena are well studied in integer-order and discrete-time models, their occurrence in ecological systems with memory remains insufficiently understood. In this work, we investigate a fractional-order tri-trophic ecological model incorporating an Allee effect. The mathematical analysis establishes the well-posedness of the system and examines the existence of equilibrium points, together with conditions for local and global stability. Criteria for Hopf bifurcation are also derived with respect to the fractional order and the

Allee-effect parameter. To explore the global dynamics, we construct high-resolution two-parameter largest Lyapunov exponent diagrams and corresponding isospoke plots. The results show that the fractional-order model preserves classical organized periodic structures, including shrimp-shaped windows, double fish-hook patterns, and periodic bands. The system also undergoes a transition from periodic behavior to chaos through Feigenbaum-type period-doubling sequences. More importantly, period-bubbling and multistability are observed, demonstrating that memory-dependent ecological systems can retain key nonlinear phenomena. Overall, ecological memory smoothens transitions between attractors but does not eliminate the fundamental nonlinear mechanisms known from classical ecological models. These findings provide insight into how memory influences stability, pattern formation, and transitions in complex ecosystems.

18. Turing Instability and Bistable Epidemic Spread on Complex Networks: Effects of Topology and Initial Seeding [C-12]

Sourav Ghorai

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Predicting whether a localized outbreak dies out or escalates into a network-wide epidemic remains a central challenge in spatial epidemiology. We develop a bistable reaction-dispersal susceptible-infected model on complex networks to examine how transmission strength, seeding location, demographic variability, and dispersal shape outbreak trajectories on Erdos-Renyi and scale-free topologies. The analysis shows that while the onset of Turing instability is primarily governed by system parameters, the resulting spatial patterns are strongly shaped by the underlying network structure. Initial infection placement is a decisive factor, with outbreaks more likely when infections originate from highly connected nodes, while peripheral seeding often fails unless a critical threshold is exceeded. We demonstrate that increasing the transmission rate or dispersal significantly lowers the epidemic threshold, facilitating large-scale outbreaks. Notably, scale-free networks display bimodal behavior, where outbreaks either remain localized or spread widely under similar conditions, reflecting structural heterogeneity. Incorporating spatial variability in recruitment rates reveals that even modest increases in mean inflow substantially enhance epidemic potential, enabling outbreaks at lower transmission rates. The study highlights the interplay between local bistability, network heterogeneity, and dynamical processes in shaping epidemic transitions, offering a framework for understanding disease spread in complex systems and designing targeted intervention strategies.

19. Immune-Driven Evolutionary Escape in Breast Cancer: A Mathematical Framework for Immunoediting and Optimal Therapeutic Control [C-31]

Anuj Kullu

Banaras Hindu University, Varanasi, India

Abstract: Breast cancer progression is governed by dynamic interactions between tumor cells and the immune system, where immune surveillance and clonal evolution jointly shape disease outcomes. We develop a six-dimensional mathematical model that captures immune-driven cancer immunoediting by distinguishing between immunogenic and immune-resistant tumor subpopulations. The model incorporates phenotypic switching from immunogenic to resistant cells, with the switching rate increasing under cytotoxic immune pressure exerted by natural killer (NK) cells and CD8⁺ T cells. We establish the positivity, boundedness, tumor-free equilibrium stability, and persistence of the proposed system. Global sensitivity analysis using Latin Hypercube Sampling (LHS) and Partial Rank Correlation Coefficient (PRCC) identifies the parameters that most strongly influence tumor burden and immune escape. Numerical simulations demonstrate that sustained immune pressure preferentially eliminates immunogenic tumor cells, thereby creating an evolutionary selection pressure that favors the expansion of immune-resistant phenotypes and ultimately drives immune escape. Optimal control analysis shows that combining immunotherapy with cytotoxic treatment effectively eliminates resistant tumor cells, substantially reduces tumor burden, and achieves long-term disease control. This framework provides quantitative insights into breast cancer immunoediting and offers a mechanistic basis for designing therapeutic strategies against tumor heterogeneity and adaptive resistance.

20. Unveiling non-linear dynamics in the discretized Hastings-Powell system: A planar study of Arnold Tongues, bell-shaped periodic windows, and multistability [C-32]

Satyabhan Singh

Banaras Hindu University, Varanasi, India

Abstract: Understanding how complex population dynamics arise from biotic and abiotic interactions is a major challenge in theoretical ecology. Discrete models offer an effective framework to capture real-world processes like seasonal breeding and generation-to-generation changes, exhibiting diverse dynamics such as stability, periodicity, chaos, and multistability. This study focuses on the discretized Hastings–Powell model to explore its nonlinear dynamics and the effects of conversion rate and handling time on system behavior. We analyze the dynamics through time-series, phase-space, one- and two-parameter bifurcation (isospoke) diagrams, and Lyapunov exponent diagrams. Notably, the isospoke diagrams reveal Arnold tongues—stable periodic structures characterized by Farey sequences—within transitional regions, and bell-shaped periodic structures characterized by Fibonacci orderings, period adding, and period doubling sequences within chaotic regimes. Overlapping periodic windows in parameter space indicate the presence of multistability. These findings highlight the rich, hierarchical organization of nonlinear dynamics in discrete-time ecological models and provide valuable insights into tri-trophic interactions.

21. Qualitative analysis and Turing patterns in a prey-predator model incorporating Allee effect and anti-predator behavior [C-33]

Kanchan Singh

Banaras Hindu University, Varanasi, India

Abstract: In this paper, we investigate a prey-predator model incorporating a strong Allee effect in the prey population and anti-predator behavior in the predator dynamics. The temporal dynamics of the system are studied through positivity, boundedness, permanence, equilibrium, and stability analyses. Conditions ensuring species coexistence are derived, and sensitivity analysis is performed to identify the parameters that most strongly influence the coexistence equilibrium. Furthermore, Hopf bifurcation analysis reveals the transition from stable coexistence to oscillatory population dynamics as the prey growth threshold and predation parameters vary. To examine the effects of spatial dispersal, the model is extended to a reaction-diffusion framework. Conditions for diffusion-driven instability are established, and a weakly nonlinear analysis is carried out to characterize pattern selection near the Turing bifurcation point. Numerical simulations in one-dimensional and two-dimensional domains support the analytical results and demonstrate the emergence of diffusion-induced stationary spatial patterns, including stripe, hexagonal, spot-like, and elongated spot structures. The results highlight the significant roles of the prey growth threshold parameter and the predation parameter in determining species persistence, temporal oscillations, and spatial organization.

22. Stability and Bifurcation Analysis of a Cancer–Immune Interaction Model under Chemotherapy [C-34]

Pinki Rani

Banaras Hindu University, Varanasi, India

Abstract: This study presents a mathematical model describing the interactions between cancer cells and immune cells in the presence of chemotherapy. The proposed nonlinear system is first analyzed to establish the positivity and boundedness of its solutions, ensuring biological feasibility. The equilibrium points of the model are then determined, and their local stability is investigated using standard dynamical systems techniques. The analysis reveals the occurrence of saddle-node, Hopf, and Bogdanov–Takens bifurcations for specific parameter values, indicating qualitative changes in the system dynamics. In particular, the Hopf bifurcation identifies the transition from a stable equilibrium to sustained oscillatory behavior. Numerical simulations and phase portraits support the theoretical findings and illustrate the influence of key parameters on the long-term dynamics of the tumor–immune system under chemotherapy. Overall, the results provide valuable insights into tumor–immune interactions and demonstrate how chemotherapy can alter disease progression and the system's long-term behavior.

23. The missing lever in Wolbachia-based mosquito control: female recruitment proportion outsmarts population replacement and population suppression [C-35]

Partha Biswas

Indian Statistical Institute, Kolkata

Abstract: Wolbachia-based population replacement is a promising strategy for controlling mosquito-borne diseases. However, most mathematical models of Wolbachia invasion focus on biological mechanisms such as cytoplasmic incompatibility, imperfect maternal transmission, and Wolbachia-induced mortality, while demographic factors like the proportion of female recruitment remain largely unexplored. This study investigates the influence of female recruitment proportion on Wolbachia invasion and dominance over wild mosquito populations through a novel three-compartment (male, uninfected female and Wolbachia-infected female) model. Treating the female recruitment proportion as the focal parameter, we analyse its effects through equilibrium analysis, invasion criteria, and bifurcation analysis. The results reveal that the female recruitment proportion plays a critical role in shaping Wolbachia dynamics, with a threshold value above which Wolbachia can successfully invade and persist in the mosquito population. A second threshold is then obtained, marking the transition to a regime in which Wolbachia-infected females become dominant over the uninfected female and male populations. Furthermore, increasing the female recruitment proportion shifts the threshold values associated with key biological factors. These shifts indicate that an increase in the female recruitment proportion enhances the resilience of the system to variations in these biological factors, thereby facilitating Wolbachia persistence and dominance.

24. Trade-off between Allee remediation and disease transmission for adaptive survival: A mathematical model on Allee-exposed species [C-36]

Mrinmay Pakhira

Indian Statistical Institute, Kolkata

Abstract: Breeding success and mate acquisition are major drivers of ecological balance. The mate-finding Allee effect (henceforth, MFA) is a serious threat to the persistence of low-density species in different taxa. Several species have already gone extinct due to the Allee effect, and many more are still affected. Now, the matter of concern is whether the remedial measure of the Allee effect will improve the species' survival portfolio. Unfortunately, limited articles address the MFA, but the remedial concept as a key model parameter has been ignored in the existing studies. When disease is present, improving mating efficiency increases transmission risk unintentionally. In existing literature, Allee remedial refers to ecological, behavioural, or management interventions. It mitigates the negative impacts of the Allee effect and improves the persistence of Allee-exposed species. Here, we define a new mathematical framework in the presence and absence of disease for Allee-exposed species. We rebuild the functional form of the Allee effect, which captures the remediation at low densities. The proposed model integrates a three-compartmental SIR model, focusing on the hypothesis of Allee remediation and how Allee remediation interacts with Allee-exposed species. Our observation reveals that Allee remediation can be a trade-off for species survival for Allee-exposed species in the presence of disease. We also find that not only Allee remediation but also crowding plays an important role in species survival. We believe this work gives in-depth knowledge about the Allee effect and its remediation, which can help ecologists and eco-epidemiologists to make decisions about their intervention strategy.

25. Stability and bifurcation analysis of an imprecise Bazykin's predator–prey model [C-37]

Dr. Bapin Mondal

Department of Mathematics, SR University, Warangal, India

Abstract: In real ecological systems, biological parameters are often uncertain due to environmental variability, limited observations, and measurement errors. In this paper, we develop an imprecise Bazykin-type predator–prey model in which uncertain biological parameters are represented by intervals rather than fixed values. The proposed model incorporates logistic growth of the prey population, a Holling Type II functional response to describe predator–prey interactions, and predator self-interaction to account for intraspecific competition. A linear parametric approach is employed to transform the interval-valued model into a suitable form that can be analyzed using standard mathematical techniques. The model is investigated in terms of its equilibrium points, local stability, and the conditions for the occurrence of different types of bifurcations. Numerical simulations with interval-valued parameters are performed to illustrate the theoretical findings and to examine how parameter uncertainty influences the qualitative dynamics of the predator–prey system. The results demonstrate that interval uncertainty can significantly affect population dynamics and may lead to changes in stability and bifurcation behavior. The proposed interval Bazykin-type framework provides a realistic mathematical approach for studying predator–prey interactions in ecological systems subject to parameter uncertainty.

26. A Delay Differential Game for Blockchain-Enabled Sustainable Supply Chains with Recycling, Trade Credit and Green technology spillover [C-43]

Animesh Manna

Sidho-Kanho-Birsha University, West Bengal, India

Abstract: Scantent industrial activity and resource consumption have pushed supply chains to move beyond pure profit maximization toward sustainability-driven operations. The escalating demand for product traceability and growing concern over customer data privacy have accelerated firms' adoption of blockchain technology. We investigate how manufacturer investment in carbon abatement and blockchain adoption jointly fosters a cleaner-circular economy while enhancing supply chain transparency. Meanwhile retailers build environmental goodwill to shape consumer demand, how recycling investment supports circular resource use. Trade credit extended to retailer, delays in technology deployment and technology spillover effects add practical real-world richness to this research.

A three-tier closed-loop supply chain, comprising a manufacturer, retailer, and recycler, is modelled through delay-differential Stackelberg game to adopt long term dynamic behaviour. Five decision regimes are analyzed: decentralized, cost-sharing, centralized, delay-free, and spillover-free. The resulting system is solved analytically before being validated through

MATLAB-based numerical simulation.

Interaction between blockchain efficiency(ϕ) and technology spillover(χ) allows the retailer's profit to exceed the manufacturer's by roughly 11%. Further analysis shows that trade credit policy and implementation delay display critical threshold effects ($\theta \approx 10.8, \tau \approx 6.5$) beyond which the profit split between manufacturer and retailer. Cost-sharing contracts have greater profit than decentralized baseline, creating win-win situation for manufacturer and retailer.

27. Analysis and Optimal Control of a Caputo Fractional-Order Ebola Model with Environmental Pathogen Transmission

[C-46]

Dipti Dolai

IEST, Shibpur

Abstract: Ebola virus disease (EVD) remains a major public health concern due to its rapid transmission and the challenges associated with effective disease containment. In this study, we develop a novel seven-compartmental SVIHRQG fractional-order model incorporating susceptible, vaccinated, infected, hospitalized, recovered, deceased, and environmental pathogen populations. The model accounts for both human-to-human and environmental transmission while incorporating memory effects through the Caputo fractional-order derivative. The qualitative dynamics of the proposed model are investigated by establishing the existence and stability of the disease-free and endemic equilibrium points. A normalized forward sensitivity analysis of the basic reproduction number is performed to identify the key parameters that most strongly influence Ebola transmission. Furthermore, a transcritical bifurcation analysis demonstrates that variations in the basic reproduction number can induce a stability switch between the disease-free and endemic equilibria. To investigate effective disease-control measures, an optimal control problem incorporating vaccination and treatment interventions is formulated and analyzed. Numerical simulations illustrate the impact of fractional-order dynamics and demonstrate that combined vaccination and treatment strategies can substantially reduce infection levels compared with individual interventions. The proposed framework provides a mathematical approach for understanding the effects of memory, environmental transmission, and control interventions on Ebola dynamics and can support the design of effective strategies for disease mitigation.

28. Mathematical and Computational Modelling of Harmful Algal Blooms: Critical Synthesis, Forecasting Challenges, and a Future Research Direction

[C-50]

Ritika Chhetri

Mizoram University

Abstract: Harmful algal blooms (HABs) arise from interacting biological, chemical, and physical processes, making reliable prediction difficult. This review critically synthesizes mathematical and computational approaches for plankton dynamics and HAB forecasting. It

focuses on ecological realism, model uncertainty, data requirements, interpretability, and computational feasibility. A quantitative analysis of selected studies published during 2015--2026 shows that model complexity is not inherently beneficial. Simple mechanistic models remain effective as interpretable baselines in well-mixed and data-limited settings. Stochastic, spatial, hydrodynamic, or hybrid formulations become more appropriate when environmental variability, transport, or heterogeneous observations strongly affect bloom dynamics. Data-driven methods can capture nonlinear patterns in high-dimensional observations, but their reliability depends on temporal non-stationarity, extrapolation, validation design, and uncertainty. Data assimilation can improve state estimation but does not remove structural model error. These findings support an incremental strategy in which each added mechanism or computational component is retained only when it provides measurable value over simpler alternatives. Based on this evidence, a five-phase framework is proposed. It integrates mechanistic modelling, multi-source observations, machine learning and physics-informed learning, explainability and causal analysis, and operational forecasting toward reliable, adaptive, and computationally feasible HAB prediction.

29. Forecasting Chaotic time-series using Nonlinear Observable Space Maps [C-51]

Raktim Ray

Senior Research Fellow, Department of Mathematics, Jadavpur University

Abstract: Forecasting chaotic time series of physical and biological systems is fundamentally challenging due to their sensitive dependence on initial conditions, and rapid growth of prediction errors. Here we introduce a nonlinear observable space forecasting framework that extracts the local geometry of the chaotic dynamics through nonlinear transformations of the delay coordinates. This method maps the delay-reconstructed states into a nonlinear observable space, where the neighbourhoods are defined using the Euclidean distances between the target vector and the other vectors in the transformed space. Local predictions are obtained using weighted regression on the corresponding observables. We have used several classes of observables, including Chebshev, Legendre, Hermite, Fourier polynomials and Gaussian RBFs to capture the geometry of the original attractor. The proposed methodology is evaluated on the Lorenz and Rössler systems, both clean and noisy, and on the Bermuda Atlantic time-series(BATS) empirical ocean data. The results show considerable improvements in the forecasting accuracy, with higher Pearson correlation and lower RMSE and MAE in comparison to standard linear maps. Particularly, the nonlinear observable representations capture the curved and localised structures of the underlying attractor in a better way and also show robustness to stochastic noise.

30. Clinical Burden of Dual Pathogens: A comparative analysis of Mono infection Versus Zoonotic co-infection outcomes using SIR method [C-52]

Shipan Chandra Deb nath

Chittagong University of Engineering and Technology, Bangladesh

Abstract: Zoonotic diseases pose a significant global health challenge, particularly when individuals are exposed to multiple pathogens simultaneously. Co-infections involving zoonotic pathogens may produce more severe clinical outcomes than mono infections due to complex pathogen–host interactions, immune system modulation, and enhanced disease transmission. This study presents a comparative mathematical analysis of the clinical burden associated with mono infection and zoonotic co-infection using a Susceptible–Infected–Recovered (SIR) modeling framework. The proposed model incorporates separate transmission dynamics for single-pathogen infection and dual-pathogen co-infection, allowing assessment of disease incidence, prevalence, recovery, and cumulative clinical burden. Key epidemiological indicators, including the basic reproduction number, infection prevalence, peak infection levels, duration of outbreaks, and total infected population, are analyzed to compare the outcomes of mono infection and co-infection scenarios. Numerical simulations are employed to investigate the influence of transmission rates, recovery rates, and co-infection interactions on disease dynamics. The model is expected to demonstrate that zoonotic co-infections can substantially increase infection burden, prolong epidemic duration, and intensify clinical outcomes compared with mono infections, particularly under conditions of synergistic pathogen interaction. The findings may provide valuable insights into the epidemiological consequences of dual-pathogen infections and support the development of integrated surveillance, prevention, and control strategies for emerging zoonotic diseases. This mathematical framework can serve as a useful tool for understanding complex co-infection dynamics and informing evidence-based public health interventions.

31. A nonlinear dynamical model of algal bloom regulation by herbivorous fish [C-54]

Prinsi Shukla

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Abstract: Algal blooms are a major ecological concern in aquatic ecosystems due to their adverse effects on dissolved oxygen dynamics and aquatic biodiversity. In this study, a nonlinear mathematical model is proposed to investigate the coupled interactions among nutrients, algae, herbivorous fish, and carnivorous fish, while incorporating the impact of algal accumulation on surface re-aeration processes. The well-posedness of the proposed system is established by proving the positivity and boundedness of solutions. The feasibility of the interior equilibrium is analyzed, revealing conditions for the existence of unique and multiple coexistence states. Furthermore, bifurcation analysis reveals the occurrence of transcritical, saddle-node, Hopf, and Bogdanov–Takens bifurcations, leading to rich dynamical behaviors such as alternative stable states, bistability, and sustained oscillations. Numerical simulations corroborate the analytical findings and demonstrate that nutrient uptake by algae and fish grazing rates play crucial roles in regulating bloom formation and ecosystem stability. The findings identify critical ecological thresholds governing regime shifts between ecologically balanced and algal bloom-dominated states, underscoring the

importance of integrating biological control with nutrient management to mitigate harmful algal blooms and promote the long-term ecological stability of freshwater ecosystems.

32. Modeling the role of media coverage and recovery awareness programs on the dynamics of infectious diseases with limited healthcare facilities [C-55]

Anoop Kumar Mishra

Indian Institute of Engineering Science and Technology, Shibpur, India

Abstract: The spread of infectious diseases in any country poses pressure on the healthcare system and also impacts the economy. In this study, we present a four-dimensional nonlinear compartmental model to control the spread of an infectious disease by considering three important control measures; namely (i) media coverage, (ii) recovery-oriented awareness programs, and (iii) hospital bed capacity. In formulating the model, we have considered that media coverage reduces the transmission rate; however, awareness programs increase the self-recovery rate due to home care and healthy practices of infected individuals, which ultimately reduces the pressure on hospitalization. For the presented model, the basic reproduction number is obtained. The feasibility of equilibria and their stability properties are discussed in detail. It is shown that under certain parameter regimes, the model exhibits interesting dynamics, which include forward and backward transcritical, saddle-node, Hopf, and Bogdanov--Takens bifurcations, leading to multistability and oscillatory dynamics. Numerical simulations indicate that higher efficacy of media coverage reduces the periodic behavior of the system, delays, and reduces epidemic peaks. The findings highlight that epidemic control requires coordinated strengthening of both media-based interventions and healthcare infrastructure.

33. A Mathematical Study of Continuous Model in Tumor–Immune Dynamics With Interleukin-2 [C-61]

Vikrant Rai

Visva-Bharati, India

Abstract: Understanding the complex interactions among tumor cells, immune responses, and cytokines, such as interleukin-2 (IL-2), is crucial for advancing cancer immunotherapy. We introduce a four-dimensional mathematical model that includes healthy cells, tumor cells, effector immune cells, and IL-2 dynamics. This study explores the continuous-time formulations of the proposed model to examine the nonlinear behaviors of tumor-immune interactions. A detailed bifurcation analysis explores the system's qualitative changes in response to variations in key immunological parameters. The analytical finding suggests the existence of Hopf and saddle-node bifurcations in the continuous-time model. We employ a data-fitting approach to estimate model parameters using real-world tumor growth data with and without IL-2 treatment. The analytical findings are validated through numerical

simulations demonstrating Hopf, saddle-node bifurcations. The findings underscore the significance of IL-2-mediated feedback in regulating immune activation and tumor suppression. This study enhances the understanding of tumor-immune-cytokine interactions and provides a computational framework for exploring stability and chaos in cancer-immune regulation.

34. A comparative investigation of the delayed and non-delayed optimal control techniques on misinformation mitigation [C-70]

Arijit Malakar

Indian Institute of Engineering Science and Technology, Shibpur, India

Abstract: The rise of online misinformation has serious and far-reaching consequences for our daily lives. It can distort our understanding of critical issues, from public health to safety, leading to misguided decisions and increased anxiety. In times of crisis, such as natural disasters or health emergencies, it becomes even more imperative for a responsible governing authority to actively monitor and regulate misleading information. Here, a delayed optimal control framework is proposed considering four groups of populations: viz., newcomers $S(t)$, hesitating-class $E(t)$, spreader-class $I(t)$, Aware-class $R(t)$. Sufficient conditions for the existence of nonnegative and bounded solutions are proved. The stability analyses of the misinformation-free state or the prevailing state are examined using the ‘Threshold value of misinformation’. Subsequently, the analysis of the delayed optimal control model is investigated using modified PMP (Pontryagin's maximum principle). Finally, and most importantly, the results obtained using the optimal control technique while accounting for the phenomenal delay are compared with those of the non-delayed system. This comparison highlights the practical limitations of misinformation-refutation techniques when the phenomenal delay is overlooked.

35. Nonlinear Dynamics of Socio-Ecological Resilience and Mental Health: An Evolutionary Game Approach [C-71]

Rana Pal

Indian Institute of Engineering Science and Technology, Shibpur

Abstract: Sustainable management of vulnerable socio-ecological systems depends on an understanding the coupled dynamics of natural resources, economic activity, adaptive capability, and mental stress. Under external environmental hazards, we develop a nonlinear mathematical model describing the interactions among available renewable resources, economic effort, adaptive governance, and accumulated psychological stress. An evolutionary game framework is incorporated in which policy approaches evolve according to their relative payoffs to capture the dynamic behavioural response of people and

institutions. The model is also extended stochastically to reflect slow changes in system parameters and random hazard events. Stability and bifurcation analyses help to pinpoint important parameter ranges and changes in system behavior, while early-warning signals based on variance and lag-1 autocorrelation are examined as indicators of changing resilience. With NDVI readings used for a qualitative comparison with simulated resource dynamics, the Indian Sundarbans are considered as a representative case study. Numerical findings show how adaptive capacity, resource extraction pressure, and cooperative governance affect resource resilience and socio-psychological stress. Key parameters affecting resilience are identified and a mathematical basis is provided in this framework for understanding adaptive policy dynamics in coupled human–environment systems exposed to multiple hazards.

36. Social dilemma analysis of an epidemic model with behavioral dynamics: An evolutionary game-theoretic approach [C-72]

Kiriti Bhusan Mahato

Indian Institute of Engineering Science and Technology, Shibpur, India

Abstract: In this study, we develop an *SuSaVITR* epidemic model that incorporates three unique behavior dynamics based on the evolutionary game theory (EGT). This study incorporates awareness, vaccination, and treatment strategies into a behavioral model, in which their adoption is driven by human behavior and risk perception, and each intervention balances its associated costs against the risk of infection. In the non-behavioral model, the normalized forward sensitivity index method is employed for local sensitivity analysis of the basic reproduction number, while the LHS-PRCC method is used for global sensitivity analysis of the infected compartment to identify the most influential parameters affecting disease dynamics. The simultaneous application of three strategies incorporating EGT with the minimum possible cost for each intervention proves to be the most effective approach for reducing the maximum number of infections compared to the application of single or double strategies. Also, the average social payoff has been evaluated at both Nash equilibrium and social optimum to emphasize the increased social benefits that emerge when the costs of awareness, vaccination and treatment are minimized. Finally, the social efficiency deficit has been discussed to identify the situations that lead to dilemmas related to the epidemic model.

37. Analysis of a non-autonomous dynamical system : Local stability and attractors [C-68]

Sagnik Bhaumik

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Environmental and complex biological phenomena in real world are understood and analysed using with the help of dynamical systems which provides a mathematical framework to describe the behaviour of these systems. Modelling a non-isolated biological event with taking the time dependent parameter into consideration can give us more accurate

outcome. So, the study of non-autonomous system becomes much more important nowadays to predict the behaviour such as critical transitions and tipping for several environmental phenomena.

Here, our work presents a study of a well-known mathematical model in non-autonomous platform which gives insights of behavioural difference between autonomous and non-autonomous systems. Along with this comparative study of autonomous and non-autonomous model, we have done both analytical and numerical study of the non-autonomous system which show how crucial the time-dependence is for any system. This work also explores the understanding of different non-autonomous attractors and their nature of attractivity. As a detailed analysis of non-autonomous system along with some paradigmatic examples, this study however helps us to understand the behaviour of a non-autonomous model which will however be important to study tipping phenomena and non-autonomous dynamics more accurately.

38. A Mathematical Model of Antidote-Induced Inhibition in Methanol Poisoning under Double Impulsive Therapy [C-69]

Sanju Sardar

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Methanol poisoning from homemade or illicit alcohol remains a neglected public health threat, especially in low- and middle-income countries. Without prompt treatment, it can cause severe metabolic acidosis, permanent visual loss, neurological damage, and death. Treatment aims to block methanol metabolism and enhance formic acid clearance. Here, we derive a Michaelis–Menten-based inhibition function to capture antidote-induced inhibition and use it to model combined ethanol and folinic acid (E–F) therapy under impulsive dosing. We obtain periodic solutions, determine the toxicity-free equilibrium, and analyze its stability using Floquet theory. An analogous model is developed for fomepizole combined with folinic acid (Z–F). Numerical simulations identify optimal fomepizole regimens, compare dosing strategies, and estimate the time needed to eliminate toxic metabolites. Finally, a cost-effectiveness analysis using ACER and ICER compares Z–F and E–F therapies, providing mechanistic and economic insight to guide clinical and public health decisions.

39. Dynamical Analysis of Sterile Insect Technique for Sustainable Pest Control in Jatropha Plantations [C-11]

SK Mosaraf Ahammed, Kunal Paul

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: The growing demand for renewable biofuels has increased interest in *Jatropha curcas* as a sustainable biodiesel source. However, large-scale plantations are threatened by

infestations of *Pempelia morosalis*, which can substantially reduce biomass yield. Conventional chemical pest-control methods also pose environmental and operational challenges. This study develops a mathematical framework to evaluate the Sterile Insect Technique (SIT) as an eco-friendly strategy for pest management in *Jatropha* plantations. Three scenarios are considered: an uncontrolled pest-infested system, continuous sterile-male release, and periodic impulsive release. The model is analyzed using Normalized Spectral Sensitivity Analysis (NSSA), equilibrium and stability analysis, Hopf bifurcation, Floquet theory, and cost-effectiveness analysis. The results identify the key parameters influencing pest persistence and establish threshold conditions for effective pest suppression. The mating rate is identified as a critical parameter associated with Hopf bifurcation, while continuous SIT can ensure global stability and pest eradication. Numerical simulations and ACER and ICER analyses further demonstrate that both SIT strategies effectively suppress pest populations and support biomass recovery, with periodic impulsive releases offering greater economic and operational feasibility. These findings provide a quantitative foundation for designing sustainable SIT programs for long-term protection of *Jatropha curcas* plantations.

❖ Theoretical Physics:

40. Exact Gravastar Solution [C-04]

Anikul Islam

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Abstract: Lindemann's theorem about the transcendence of numbers and in the context of Field theory result that every constructible number is algebraic over $\mathbb{Q}$ suggest that 1 radian is inconstructible by ruler and compass. Therefore, our construction of 1 radian with ruler and compass precisely with a minimal deviation of 0.00000009 ($9/10^8$) radian is mentionworthy.

41. On Conformal Fractional Ricci Flow [C-07]

Anisha Saha

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: In this paper, we integrate the conformal Ricci flow through Perelman's functionals within a fractional and non-holonomic framework. We study the monotonicity properties of a class of fractional frequency functional associated with the conformal Ricci flow defined on a closed Riemannian manifold of constant curvature and dimension $n \geq 3$. The analysis is carried out using fractional calculus, particularly Caputo and Riemann-Liouville derivatives and integrals, to extend classical monotonicity results to the fractional setting. Furthermore, we establish several consequences of these monotonicity properties and discuss their implications within fractional geometric framework.

42. On Cauchy Split Continuous Functions

[C-08]

Sujan Khna

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Abstract: In this work, we investigate the notion of Cauchy split continuity, a weaker form of continuity introduced in connection with split continuous functions. We first establish a sequential characterization of Cauchy split continuity in terms of the images of Cauchy sequences and the associated split function. We then characterize the possible behavior of the image of a Cauchy sequence under a Cauchy split continuous function, showing that it is either Cauchy or can be decomposed into exactly two non-asymptotic Cauchy subsequences. Several relationships between Cauchy split continuity, split continuity, uniform split continuity, and Cauchy regularity are discussed through examples. We further show that total boundedness of the domain is characterized by the boundedness of images under Cauchy split continuous functions. A weaker notion, namely weakly Cauchy split continuity, is introduced and its relation with split continuity is studied, particularly under completeness assumptions. Finally, we investigate extension properties of Cauchy split continuous functions to the Cauchy completion of the domain. These results provide a broader understanding of Cauchy split continuity and its role in the study of generalized forms of continuity.

43. Study of Sasakian manifolds admitting hyperbolic Ricci soliton with respect to Zamkovoy connection

[C-15]

Nilofer Ali

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Abstract: In this paper, we studied some properties of the ϕ -conformally flat and ξ -conformally flat Sasakian manifolds with respect to the Zamkovoy connection. After that, we investigate the nature of hyperbolic Ricci soliton on Sasakian manifolds with respect to Zamkovoy connection when the potential vector fields are Reeb vector field, torse forming vector field, concircular vector field, holomorphically planar conformal vector field, etc. Finally, an example of a Sasakian manifold is presented which satisfies a result of this paper.

44. Dynamics of particles around weakly magnetized Riessner Nordstrom black hole in presence of dark energy

[C-17]

Prosanta Mandal

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Oncolytic virotherapy offers a targeted cancer treatment by selectively infecting and destroying malignant cells while preserving healthy tissue. In this study, we develop a nonlinear impulsive mathematical model to analyze interactions between uninfected and infected cancer cells (UCC and ICC) under periodic oncolytic virotherapy. The model

incorporates logistic tumor growth, nonlinear cell-to-cell transmission, and impulsive conversion of susceptible cells, providing a biologically realistic yet analytically tractable framework. We derive an explicit threshold for the dosing interval that determines treatment outcome: below this threshold, the cancer-free periodic solution is globally asymptotically stable, ensuring tumor eradication, whereas above it, the system exhibits persistence. Numerical simulations and bifurcation analysis reveal a rich range of dynamics, including periodic, multi-periodic, and chaotic oscillations driven by treatment frequency and infection efficiency. These results highlight that therapeutic success depends on the interplay between dosing schedule and viral effectiveness. The study provides new insights into tumor-virus dynamics and offers a theoretical basis for optimizing oncolytic virotherapy protocols.

45. Study of Single-Step and Multistep Numerical Methods for the Two-Body Satellite Orbit Problem [C-44]

Titli Biswas

Defence Institute of Advanced Technology, India

Abstract: The two-body problem is a fundamental model in astrodynamics for determining the motion of a satellite under the gravitational influence of a central celestial body. In this work, a systematic numerical investigation of the two-body satellite orbit problem is presented using single-step and multistep numerical integration techniques. The governing second-order nonlinear ordinary differential equations describing the satellite's position are reformulated as an equivalent system of first-order ordinary differential equations and solved numerically over a prescribed time interval. The theoretical order of accuracy, consistency, convergence, and error characteristics of the numerical schemes are examined to establish their suitability for orbital propagation. Numerical experiments are conducted for representative initial orbital conditions, and the computed satellite position and velocity are used to reconstruct the corresponding orbital trajectories. The numerical solutions obtained using different methods. Particular emphasis is placed on examining the effect of the order of the numerical method and integration step size on the accuracy of satellite position and velocity over the propagation interval. The study offers a numerical framework for selecting appropriate integration schemes for reliable and accurate prediction of satellite trajectories in the two-body gravitational model

❖ Solid Mechanics:

46.Guided wave propagation under the influences of magnetic field and initial stress on in thin generalized magneto-thermoelastic plate with two temperatures [C-39]

Zenith Oinam

Mizoram University, India

Abstract: The problem of propagation of elastic waves in a generalized magneto-thermoelastic plate has been investigated. The effects of electromagnetism and initial stress have been examined within the context of the Green–Naghdi thermoelastic theory of type III (GN-III) with two temperatures. The dispersion relations for symmetric and antisymmetric vibrations are derived, and the phase velocities and attenuation magnitudes are numerically computed. The combined effects of two temperature parameter, magnetic field and initial stress are found to have significant impact on wave dynamics. Additionally, the thickness of the plate considerably influence the wave characteristics. From the current formulation, we have also reduced to the problems of thermoelastic theory, *GN – II* and classical elasticity.

47. Reflection phenomena in a visco-thermo-micropolar elastic medium with the micro-temperature.

[C-57]

Arambam Bickyrose Chanu

Mizoram University, India

Abstract: We have investigated the problem of reflection of elastic waves in a visco-thermo-micropolar elastic medium. We have derived the phase velocity of longitudinal, transverse, micropolar and temperatures fields. We have observed that there are three coupled longitudinal waves and three coupled transverse waves. Under the context of Lord-Shulman thermoelastic theory, we have derived and computed the amplitude and energy ratios of the reflected waves due to incident longitudinal/transverse waves using appropriate boundary conditions. Using MATLAB software, these ratios have been numerically computed and the results have been analysed.

48. Propagation of Rayleigh wave in anisotropic thermo-piezoelectric Material

[C-62]

Lalruatpuii Colney

Mizoram University, India

Abstract: The paper investigates the propagation of Rayleigh surface waves in a homogeneous transversely isotropic thermo-piezoelectric medium. A normal mode analysis is employed to reduce the governing partial differential equations to a system of ordinary differential equations, which is then formulated as a first-order eigenvalue problem. Eigenvalue approach is used to determine the admissible partial-wave modes and their corresponding attenuation characteristics. Using appropriate mechanical, electrical, and thermal boundary conditions at the free surface, a secular equation governing the Rayleigh-wave propagation is obtained. Numerical results are presented to examine the dispersion and attenuation characteristics of Rayleigh waves and to demonstrate the influence of the coupled thermo-piezoelectric parameters on surface-wave propagation.

49. Spillover-Free Finite Element Model Updating with Incomplete Spectral Data: An Optimization Approach [C-49]

H. Lalhriatpuia

Mizoram University, India

Abstract: This study presents a systematic approach for updating symmetric structured finite element models using partial spectral information, where selected eigenvalues and specific entries of the corresponding eigenvectors are prescribed. The proposed approach preserves the symmetry and structural properties of the finite element model, eliminates spillover effects, and minimizes perturbations to the mass and stiffness matrices. Necessary and sufficient conditions for the existence of solutions are established, and the complete set of admissible solutions is characterized. The resulting finite element model updating problem is formulated as a constrained optimization problem that seeks minimal structural modifications while satisfying the prescribed spectral constraints and maintaining the no-spillover property. Numerical examples demonstrate the effectiveness and robustness of the proposed approach in updating symmetric structured finite element models from incomplete spectral data

50. Magnetoelastic Impact Response of a Rigid Disk at an Isotropic–Anisotropic Interface [C-67]

Dr. Subhash Chandra Mondal

Purash Kanpur Haridas Nandi Mahavidyalaya, India

Abstract: This study presents an analytical model for the transient torsional response of a rigid circular disk embedded at the interface of two dissimilar half-spaces, one of which is isotropic and the other is transversely isotropic, under the influence of an external magnetic field. The analysis extends beyond static conditions to examine the dynamic effects of impulsive torsional loading, which generates stress waves that propagate through the bimaterial medium. Magnetoelastic coupling is incorporated into the governing equations for both axisymmetric and non-axisymmetric motions, and the resulting boundary-value problem is solved using Laplace and Fourier–Hankel transform techniques, leading to a Fredholm integral equation. The developed formulation enables the determination of the transient interfacial shear stress and the dynamic stress intensity factor (DSIF) at the disk edge. Numerical results demonstrate that the mismatch in material properties significantly affects stress concentration levels, wave reflection behavior, and the temporal evolution of the DSIF, while the applied magnetic field alters the dynamic response through magnetoelastic interactions. The proposed framework provides valuable insight into fracture initiation and interfacial debonding in layered and composite structures subjected to high-rate torsional loading.

51. Diffraction of Love Waves by a Rigid Strip at the Interface of an Orthotropic Layer and an Isotropic Half-Space [C-66]

Sandip Mistry

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Love waves (Q-waves in seismology) are fundamental guided waves in layered elastic media. This study examines the scattering of an incident fundamental Love mode by a thin rigid strip inclusion at the interface between an orthotropic elastic layer and an isotropic half-space, where the layer acts as a waveguide. The dispersion relation and phase velocity follow from the boundary conditions of interfacial Love-wave propagation. Using the Fourier transform, the mixed boundary-value problem is reduced to a Cauchy-type singular integral equation whose kernel is evaluated by contour integration. The equation, closed by the displacement condition on the strip, is solved by Gauss–Chebyshev collocation, and the dynamic stress intensity factor (DSIF) at both edges is obtained by Krenk's extrapolation formulae. Numerical results for a graphite/epoxy layer on steel show how the DSIF depends on the strip width, the layer thickness and the wave frequency, and that the edge facing the incident wave generally carries the larger stress intensity. The findings are relevant to nondestructive evaluation, seismic modelling and structural health monitoring of composite materials.

❖ Fluid Dynamics

52. Mixed Convective Heat Transfer and Entropy Generation in an Inclined Non-Square Cavity with a Power-Law Nanofluid [C-13]

Prasanta Mandal

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: This study investigates laminar conjugate mixed convection of a power-law nanofluid in an inclined non-square cavity comprising a conductive lower wall, a cold moving upper lid, and adiabatic inclined side walls. Nanoparticle transport is modeled using a non-homogeneous formulation, incorporating the influence of local nanoparticle concentration on the effective thermophysical properties of the suspension. The coupled governing equations for fluid flow, heat transfer, and nanoparticle concentration are formulated in dimensionless form and solved using a radial basis function (RBF)-based meshless method, in which computational centers are distributed directly within the physical domain, eliminating the need for body-fitted mesh generation. The numerical results are further validated against available theoretical and experimental data. The effects of cavity inclination, side-wall inclination, wall thermal conductivity, power-law index, and nanoparticle migration on heat transfer and entropy generation are examined using the average Nusselt number, entropy generation, and Bejan number. The results provide insights into the thermal and thermodynamic characteristics of inclined-cavity systems relevant to solar thermal collectors, electronic cooling, thermal energy storage, and food-processing applications.

53. Numerical Investigation of Fluid Mixing in Passive Microchannels with Varying Cross-Sections [C-24]

Sharif Mahmud Bhuiyan

Indian Institute of Information Technology Kalyani

Abstract: Micromixers are one of the vital components of micro fluidic devices since many biological and chemical applications require homogenous mixing. Recent research has increasingly paying attention on optimizing micromixer designs, materials, and fabrication methods to deliver high-performance devices for targeted biomedical and chemical applications. At the micro scale, the flows are laminar because of particularly low Reynolds numbers, resulting in weak convective mixing and dependence on slow molecular diffusion. This study represents a numerical investigation of fluid mixing in passive microchannels in varying cross- sections. A three-dimensional computational model is developed to analyze the influence of geometric variations on flow behavior and mixing performance. The governing continuity, Navier-Stokes, and advection-diffusion equations were solved via the finite element method (FEM). The effects of Reynolds number, channel geometry, and cross-sectional variation in mixing efficiency, and flow patterns are examined. The results demonstrate that varying cross-sections induce fluid stretching, and enhanced interfacial contact between fluid streams, leading to improved mixing compared to conventional straight microchannels. The finding will provide valuable insights for the design of energy-efficient passive micromixers in biomedical, chemical and lab-on-a-chip (LOC) applications.
Keywords: Computational fluid dynamics (CFD), Micro fluidics, Numerical simulation, fluid mixing, Passive micromixer, varying cross section, mixing efficiency

54. Numerical Investigation of Mass Transfer Dynamics in Mango Slices during Osmotic Dehydration Using FEM [C-27]

Ripa Muhury

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Abstract: Fruits and vegetables are rich sources of nutrients but deteriorate rapidly due to their high moisture content, which limits their shelf life and leads to postharvest losses. Osmotic dehydration is an efficient non-thermal preservation technique that extracts water content while preserving the nutritional and sensory attributes of food products. This study constructed a 2D axisymmetric FEM in COMSOL Multiphysics to predict the osmotic dehydration of mango slices in 5% stevia solution at 50 °C. This study employed the transport of diluted species interface to simultaneously predict water loss and solute gain by solving coupled diffusion equations within the mango tissue and osmotic solution. Water and stevia transport were characterized using concentration-dependent diffusion coefficients and appropriate initial and boundary conditions. Simulation results demonstrate a rapid initial water loss driven by the high osmotic gradient, followed by a gradual reduction in the dehydration rate as the system approaches equilibrium. Concurrently, stevia diffused into the

mango tissue, resulting in progressive solute uptake and non-uniform concentration distributions across the slice thickness.

55. Thermosolutal Convection of a radiative ternary hybrid nanofluid under an applied magnetic field in a porous cavity with half-sinusoidal heating [C-45]

Dhruba Majhi

Sidho-Kanho-Birsha University, WB, India

Abstract: This work emphasizes the role of the nonuniform multi-frequency heating and concentrating on irreversibility analysis and thermosolutal convection phenomena of a radiative ternary hybrid nanofluid in a curved porous container. A numerical analysis is conducted to analyze thermal and species convection within a convex permeable enclosure packed with a radiative aqueous ternary hybrid nanoliquid, with particular emphasis on the role of the nonuniform multi-frequency heating and concentrating. This geometry offers unique opportunities and challenges for improving thermal management and efficiency across various applications, including heat exchangers, solar-powered collectors, nuclear magnetic resonance imaging, electronic cooling, chemical reactors, and processing units. Adopting a Higher-Order Compact (HOC) numerical scheme, to overcome these challenges, the governing Navier-Stokes equations, together with the temperature and species equation, are determined. This research explores the effect of numerous important parameters, such as the Lewis number, Darcy number, buoyancy ratio number, radiation parameter, Rayleigh number, frequency, amplitude, offset temperature of the sinusoidal heating, and trihybrid nanofluid's solid particle concentration. The results highlight that a structure of multicellular flow within the chamber is generated by an increase in the sinusoidal frequency of the bottom surface temperature.

56. Electrophoresis of Polyelectrolyte-Adsorbed Soft Particles with pH-Regulated Charge Densities of the Inner Core and Shell [C-47]

Asim Mahata

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: This article investigates the electrophoresis of a polyelectrolyte-adsorbed soft particle in a binary electrolyte solution. The peripheral polyelectrolyte layer (PEL) contains pH-regulated zwitterionic functional groups that give rise to a pH-dependent volume charge density. The monomer density and the corresponding volume charge distribution within the PEL are modeled using an exponential decay function based on the profile of the adsorbed polymer. Additionally, the hydrophobic inner core bears zwitterionic surface groups, yielding a pH-regulated core surface charge. Taking these coupled electrokinetic and surface properties into account, we model the electrophoretic motion of the soft particle. Based on a flat-plate approximation—valid when the particle radius well exceeds the electric double layer thickness—we derive an explicit, closed-form expression for electrophoretic mobility within the Debye–Hückel electrostatic framework using the homotopy perturbation method.

We further extend our analysis to encompass moderately to highly charged soft particles. The results for electrophoretic mobility are presented to illustrate the influence of key governing parameters on the particle motion. The results derived in this work provide a foundation for more refined calculations of the electrokinetic behavior of pH-responsive, polyelectrolyte-adsorbed soft particles

57. Multiscale pattern complexity in dry-land vegetation: A simplified modeling approach incorporating root augmentation and stratified soil water dynamics [C-53]

Sounov Marick

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Abstract: Arid and semi-arid ecosystems frequently exhibit striking self-organized vegetation patterns that emerge as collective adaptations to water scarcity. Although reaction–diffusion models have successfully reproduced many of these spatial structures, the role of plant-level adaptive traits, particularly root phenotypic plasticity, remains insufficiently understood. In this study, we present a minimal reaction–diffusion model that incorporates vertical soil stratification and adaptive root conversion to investigate the emergence of multiscale vegetation patterns. Extending the classical framework of Hardenberg et al. (2001), we introduce a two-layer soil-water system and a unidirectional transition from shallow-rooted to deep-rooted plant phenotypes. Through analytical and numerical investigations, we demonstrate that strong contrasts in diffusion rates, together with adaptive root conversion, promote the coexistence of two distinct Turing instabilities. These interacting modes generate robust multiscale vegetation structures, where highly diffusive shallow-rooted biomass forms broad spatial bands, while less diffusive deep-rooted biomass organizes into finer-scale clusters. Our results provide a mechanistically simple yet ecologically realistic explanation for multiscale pattern formation in drylands, highlighting how adaptive root strategies interact with coupled soil-water dynamics to shape ecosystem spatial organization.

58. Modeling the role of aerial water spraying in combating atmospheric pollutants and sustain industrialization [C-56]

Gauri Agrawal

Amity University Uttar Pradesh, Lucknow Campus

Abstract: In the pursuit of global development and civilization, the accelerated pace of industrialization has brought significant atmospheric challenges, posing serious risks to public health. This study examines the implications of industrial establishment and relocation/ closure, focusing on the emission of atmospheric pollution, all associated with the human population. A novel mathematical model is formulated, which reveals complex dynamical behaviors, including transcritical, saddle-node, subcritical, and supercritical Hopf

bifurcations. Critical thresholds are identified for the establishment rate (and relocation/closure rate) of industries, beyond which the system loses (and gains) its stability. To overcome with the anthropogenic atmospheric pollution, the model system is extended by introducing a new dynamic variable, aerial water spraying, as a pollution removal technique. Extended system significantly suppresses bifurcations, terminates limit cycle oscillations, and promotes asymptotic stability. Further findings reveal that increasing the water spraying rate reduces the concentration of atmospheric pollutants and simultaneously increases the densities of human population and industries. The system's stability region expands as the spraying rate increases, along with the increasing industrial establishment rate. Also, the concentration of atmospheric pollutants attains a minimum when both the aerial water spraying rate and the scavenging rate of atmospheric pollution through aerial water spraying are maximized.

59. Multi-Source BOD–DO Dispersion Model with Variable Hydraulics [C-59]

Jenny Lallawmzuali

Mizoram University, India

Abstract: Mathematical models describing biochemical oxygen demand (BOD) and dissolved oxygen (DO) dynamics are important instruments for assessing river water quality and supporting pollution management. In eastern India, the Damodar River basin, particularly the heavily polluted Dhanbad coal belt, receives untreated or partially treated discharges from underground and open-pit coal mines, coal washeries, thermal power plants, and heavy manufacturing units. This paper develops a multi-source BOD–DO dispersion model for heavily contaminated river systems with spatially variable hydraulic conditions. The model retains the longitudinal dispersion term within the advection–dispersion equation (ADE) and represents the flow velocity through a power-law, piecewise-linear spatially varying function $u(x)$, together with a dynamic, thermally corrected reaeration coefficient $k_2(x)$. A toxicity correction factor is incorporated into the deoxygenation coefficient k_1 to account for the inhibitory effects of industrial pollutants on microbial degradation kinetics. The resulting formulation provides a computationally efficient and mathematically rigorous tool for environmental engineering management, enabling environmental protection agencies to move from blanket effluent standards toward targeted, localized load allocations and combined toxic-organic remediation strategies.

60. Double-diffusive convection of a Casson tri-hybrid nanofluid in a plus-shaped enclosure with a baffle [C-63]

Sarna Soren

Department of Mathematics, NIT Durgapur

Abstract: In this work, a plus-shaped geometrical configuration is considered, for which limited research has been reported in the literature, especially in the presence of double-diffusive convection, Casson tri-hybrid nanoparticle effects, and a plus baffle. The purpose of this work is to investigate double-diffusive convection of a Casson tri-hybrid

nanofluid in a plus-shaped enclosure with a baffle. The governing equations describing continuity, momentum, energy, and solutal transfer are solved using a finite difference method. To analyze the influence of key controlling several parameters on fluid motion, thermal, and species transport characteristics. The results indicate that the centrally located cold plus-shaped obstruction significantly reorganizes the flow patterns and thermal fields, leading to notable enhancement in transport performance. The results indicate that the inclusion of a Casson tri-hybrid nanofluid, a plus baffle, and a plus-shaped enclosure significantly enhancement in the overall thermal transport performance. This work fills the identified research gap by providing the first integrated analysis that simultaneously considers Casson rheology, ternary hybrid nanoparticle effects, double-diffusive transport and obstruction-induced flow modification within a plus-shaped enclosure. These findings have practical relevance across many engineering applications, including electronic cooling systems, compact heat exchangers, battery thermal management units, and other space-constrained thermal applications.

61. Adaptive Grid Refinement Framework For Air Quality Dispersion Modeling In Opencast Coal Mines [C-64]

K.Lalnunsiami

Mizoram University, India

Abstract: High resolution concentration profile is of great importance for air quality prediction in opencast coal mines due to their importance in the assessment of environmental impacts and ensuring regulatory compliance. Coarser uniform grids often failed to capture accurate concentrations intensity while finer uniform grids demand immense computational loads. Thus, this paper presents an adaptive grid refinement method by evaluating the spatial coordinates of mine source geometries specifically for opencast coal mines. The framework uses the techniques that increase grids density near high impact zones and coarsening the grids near low impact zones and hence reduces the total receptor counts by increasing resolutions only in the crucial area. The proposed computational framework when applied in an opencast coal mine showed accurate exposure assessments in areas of interest while still maintaining computational efficiency. This computational framework provides high resolution concentration profile of pollutants but still computationally lightweight which may serve to be useful for environmental regulators for environmental impact assessments and for the formulation of mitigation strategies.

62. A Wind-Oriented Grid-Based Mathematical Framework for Efficient Source-Receptor Analysis of Particulate Dispersion in Open-Cast Coal Mines [C-65]

Lal Prakash Baral

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Abstract: Suspended particulate matter (SPM) generated by open-cast coal-mining activities is an important environmental concern, with its spatial distribution strongly influenced by emission sources and meteorological conditions, particularly wind direction. Conventional

spatial discretization commonly employs fixed Cartesian computational grids whose orientation remains unchanged as atmospheric transport direction varies. This can complicate source–receptor analysis and lead to unnecessary processing of spatial regions with limited directional relevance. This paper presents a wind-oriented grid-based mathematical framework for directional source–receptor analysis of particulate emissions in open-cast coal-mining environments. The methodology employs a receptor-centred coordinate system, wind-direction-dependent coordinate transformation, longitudinal and transverse coordinates, and a directional influence mask to identify spatially relevant regions. The proposed transformation preserves Euclidean distances while aligning the computational coordinate system with the prevailing wind direction. The framework accommodates discrete and spatially distributed emission sources and can be updated for time-dependent meteorological conditions. The method operates as a preprocessing layer for established atmospheric dispersion models, including Gaussian plume and puff-based approaches. Its linear computational complexity makes it suitable for high-resolution spatial domains. The framework provides a simple, model-independent approach for incorporating changing wind direction into particulate-dispersion modelling and improving directional source–receptor analysis.

❖ Operations Research (O-R)

63. Fairness and Power in Sustainable Supply Chain: A Differential Game with Blockchain-Enabled Coordination [C-05]

Subhamoy Bera

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Abstract: The transition toward smart and sustainable supply chains requires firms to balance economic efficiency, environmental responsibility, and transparent product information while managing conflicts arising from unequal power and profit expectations. This study develops a dynamic decision-making framework for a two-tier smart sustainable supply chain comprising a manufacturer and a retailer, where overall sustainability evolves through environmental sustainability and blockchain-enabled product traceability. A continuous-time differential game is formulated to investigate long-run strategic behavior under centralized decision-making and three decentralized power structures: retailer Stackelberg, Nash, and manufacturer Stackelberg. Nash bargaining fairness concerns and bargaining power are incorporated to examine their effects on pricing, sustainability investment, traceability, demand, and profitability. Government intervention is modeled through targeted subsidies for environmental innovation and blockchain-based traceability. Furthermore, a blockchain-enabled smart revenue- and cost-sharing contract is proposed to coordinate decentralized decisions. Results show that power distribution and fairness preferences significantly influence dynamic sustainability and economic outcomes. Higher fairness concerns tend to reduce environmental and traceability investments, weakening demand and overall sustainability, whereas government subsidies stimulate investment, traceability, sustainability, demand, and profitability. The proposed smart contract further

aligns decentralized incentives with coordinated outcomes, providing insights for designing sustainable and digitally traceable supply chains.

64. Blockchain-Based Information Sharing and Transparency Decisions in CSR-Driven Supply Chains [C-09]

Pritam Majumdar

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Information asymmetry can significantly influence transparency, sustainability investment, and coordination when supply-chain members possess distinct private information. This study examines a CSR-driven sustainable supply chain with bilateral information asymmetry, where the manufacturer privately knows its product greenness while the retailer possesses private demand information. We investigate how blockchain-enabled transparency reshapes strategic information disclosure and whether greater transparency necessarily benefits both parties. A signaling-game framework compares three information regimes: no blockchain (N), where the manufacturer signals its type through wholesale price and observable CSR investment; partial blockchain transparency (BN), where the manufacturer's green type becomes verifiable while demand information remains private; and full blockchain transparency (BS), where both parties share their private information. Perfect Bayesian Equilibrium is employed to characterize signaling behavior, followed by analytical and numerical comparisons. Results show that transparency preferences depend critically on market conditions and implementation costs. Full demand sharing benefits the manufacturer when realized market potential exceeds its expected level, whereas partial transparency can be preferable under unfavorable conditions. The retailer's willingness to share demand information increases when blockchain adoption costs are low, while higher production costs discourage blockchain participation and information sharing. Thus, full transparency is not universally optimal for both supply-chain members.

65. Do Trade-Old-for-Cash and Trade-Old-for-New policies for end-of-life products benefit industries producing innovative eco-friendly products? [C-22]

Aditya Bag

Sidho-Kanho-Birsha University, Purulia, India

Abstract: As days goes, the demand of vehicles increases, while availability of raw materials decreases. As a result, raw materials and fuel. In addition, fuel-based vehicles emit carbon dioxide continuously. As a result, environmental sustainability decreases. Addressing it, industry tries to find alternative ways so that it conserves raw material, which are used for build vehicles, saves fuel, maintains sustainability. Motivated from, this paper develops a stochastic-demand closed-loop supply chain model in which eco-friendly products are manufactured using recovered materials obtained from end-of-life products. The proposed model incorporates two product return strategies, Trade-Old-for-Cash and Trade-Old-for-New, motivate customers to return old products. In addition, shortage and

refund policies are considered to investigate their effects on supply chain performance. The objective is to increase the industry's profit while ensuring resource conservation, promoting green innovation in production, and maintaining environmental sustainability. The formulated maximization problem is optimized, and several special cases are analyzed to compare economic benefits. A numerical study is conducted to illustrate the applicability of the proposed model, analyze several cases, and validate the theoretical findings. Furthermore, sensitivity analysis is performed to examine the impact of key parameters on the optimal decisions. The analysis reveals that applicable of trade-old-for-cash, trade-old-for-new has a positive impact, and the refund policy negatively affects the industry's profitability under the considered settings. The proposed model useful insights for industries into the implementation of trade-old-for-cash, trade-old-for-new programs, shortage management, and sustainable closed-loop supply chain operations

66. Optimal Replenishment and Shipment Decisions for a Manufacturer–Retailer System under Trade Credit and Cost Uncertainty [C-42]

Shantanu Mahato

Sidho-Kanho-Birsha University, Purulia, India

Abstract: Trade credit is widely used by manufacturers to encourage retailer demand, but coordinating replenishment and shipment decisions becomes more complex when cost parameters are imprecise. This paper proposes a coordinated two-echelon inventory model where a manufacturer, producing at a finite rate, supplies a retailer through multiple equal shipments per cycle and offers a fixed credit period before payment is due. Optimal expressions for the replenishment cycle and shipment frequency are derived for three cases based on the relationship between cycle length and credit period, with convexity confirmed to ensure a unique solution. Since costs such as ordering, setup, transportation, procurement, and production are often uncertain, six parameters are modeled as triangular fuzzy numbers and solved using α -cut-based parametric nonlinear programming, then converted to a crisp value via signed-distance defuzzification. Numerical results show an optimal cycle of 0.32 years, two shipments per cycle, and an annual cost of about Rs. 15,693, with the fuzzy solution converging to this value as confidence level increases. Sensitivity analysis reveals that longer credit periods reduce total cost and increase order quantity while keeping shipment frequency stable, and that interest rates affect total cost more than the retailer's ordering cost does.

67. Game-theoretic framework for centralised and decentralised supply chain coordination under imperfect production and trade credit [C-60]

Shruti Mahato

Sidho-Kanho-Birsha University, Purulia, India

Abstract: Along with supply chain complexity, successful maintenance of vendor-buyer relationship over the years has become a major challenge, though it is essential for achieving operational efficiency and long-term profitability. To tackle it, this study proposed a single vendor multi buyer integrated inventory model with planned shortages with complete backlogging. To boost larger orders and overall performance, the manufacturer provides trade credit incentives to the retailers who place orders above a specified threshold. On buyers' side, the received items go through a complete screening and defective items are returned to the vendor's warehouse. Consumer demand is sensitive to price. Lead time and defect rates are accounted as stochastic variables.

The formulated framework then compares centralized and decentralized decision-making approaches, capturing strategic interactions of players. Stochastic genetic algorithm (GA) is applied to solve the models. Results approve, in centralized scenario, vendor and buyers earn 4.24% and 3.6-4.01% more and the number of batch size is also reduced, helping to gain nearly 3.89% profit savings overall. Moreover, planned backorder, despite their costs, can increase profitability, describing the necessity of coordinated decision-making. Variation in outcomes with changing parameters is captured through sensitivity analyses. Furthermore, important managerial guidelines and future research scopes are discussed.

❖ Algorithm, Statistics & other interdisciplinary Science

68. Models on Tramadol and Polysubstance Use

[C-48]

Nivedita Singh

Mizoram University, India

Abstract: Substance use and dependence are important public health concerns that have been studied using mathematical models. The study presents a comparative analysis of mathematical models developed for heroin, alcohol, illicit drugs, prescription opioids, tramadol and polysubstance use. Models are classified according to their compartmental structures, transition mechanisms, intervention pathways and social interactions. The study extends the Multiple Relapse Tramadol Abuse Model (MRTAM) by including law enforcement as a control parameter that reduces transitions associated with drug misuse. The mathematical properties of the model are established by proving existence and uniqueness of solutions using the Picard Lindelöf theorem, with positivity and boundedness of the solutions. The study identifies important differences in modeling substance misuse by comparing the results with existing models of heroin, alcohol, opioids, and polysubstance use. The findings provide a mathematical basis for understanding tramadol misuse and to assess suitable intervention strategies.

69. Quantifying Explanation Shift in Deep Neural Networks: A Mathematical Modelling Framework for Adversarial Detection [C-26]

Avishek Chowdhury

Chittagong University of Engineering and Technology, Bangladesh

Abstract: State-of-the-art Convolutional Neural Networks exhibit a well-documented vulnerability to visually imperceptible adversarial perturbations. These minimal, mathematically bounded alterations to input spaces induce severe output misclassification and compromise model integrity. While Explainable Artificial Intelligence (XAI) methods provide transparency by linking input features to predictive relevance, adversarial attacks invariably distort these explanatory mappings. This phenomenon, termed “Explanation Shift,” lacks rigorous mathematical formalisation as a detection mechanism. In this paper, a comprehensive quantitative framework is proposed to model explanation shift as a statistically measurable signature of adversarial tampering. By treating XAI-generated heatmaps as both high-dimensional vectors and spatial probability distributions, a suite of mathematical distance and similarity metrics is applied to quantify divergence in the explanation space. Using a ResNet architecture trained on the CIFAR-10 benchmark, the model is subjected to diverse gradient- and optimisation-based attacks (under L_∞ and L_2 norms) and the resulting explanation shifts are computed. Our comparative analysis identifies the optimal functional pairing of XAI methodologies and mathematical distance metrics that maximises linear separability between clean and adversarial samples. Ultimately, this work demonstrates that applying precise topological and vector-based distance metrics to model internal reasoning mappings provides a highly robust, theoretically grounded foundation for interpretable anomaly detection in computer vision.

70. Certified Stability Radii and Variance-Based Sensitivity of a Fuzzy AHP Soil Health Index: Analytical Results and Validation on Acidic Hill Soils of Mizoram, India [C-70]

Laldinpuia

Mizoram University, India

Abstract: Fuzzy multi-criteria soil health indices are widely used, yet the stability of their class labels is rarely analyzed. We study the SHDS-FAHP-QI index, a Chang fuzzy AHP weighted sum of trapezoidal scoring functions over eight soil indicators. We prove that the index is fully compensatory, so any single indicator can fail completely while a site stays Excellent. We derive closed-form certified radii for class stability under weight perturbation in the l_1 and l_2 norms, an exact path for one-at-a-time weight changes, a Lipschitz bound under measurement error and analytic first-order Sobol indices that equal the total indices. On 53 field sites from Phura, Mizoram, 47 sites keep their class for any l_1 weight shift below

0.10 and 45 are certified against realistic sensor error, with no violation in Monte Carlo trials. On a 1000-sample synthetic set, global sensitivity ranks nitrogen and potassium above pH although pH carries the largest weight. Every index value can thus carry a computable stability tag.

71. Early Detection of Sudden Transitions in cell-cell communication in Notch Signalling [C-71]

Baishakhi Tikader

Indian Institute of Science Bengaluru, India

Abstract:

Identifying sudden transitions during phenotypic decision-making of complex biological systems is crucial for our ability to control cell states. Yet prior determination of these sudden transitions or tipping points remains challenging, as biological systems often exhibit only subtle early changes, which are often masked by inherent noise or rapid transition dynamics. Using Notch signalling as a model, we systematically analyse dynamical transitions in Notch–Delta (ND), Notch–Delta–Jagged (NDJ) and Fringe- mediated NDJ systems for both one- and two-cell contexts. In the one-cell ND system, critical slowing down-based early warning signals reliably capture transitions between sender (S) and receiver (R) states and remain robust to variations in forcing rate. We further find that flickering is a precursor to transitions in the one-cell NDJ system. By contrast, flickering does not occur in the two-cell Notch model owing to the presence of a supercritical bifurcation. Our analysis also offers insight into how Notch intracellular domain- driven and Fringe-mediated asymmetries, along with the strength of external signals, control the emergence of flickering. Moreover, the stochastic potential analysis and the bimodal distribution quantitatively measure the existence of flickering. Overall, this study identifies sudden transitions in Notch signalling under demographic noise and can be extended to other noisy biological systems.

72. Dengue Epidemic Dynamics in an Imprecise Environment: Mathematical Analysis and Optimal Control [C-72]

Gayatri Roy

Jadavpur University, kolkata

Abstract: Dengue remains a major global public health challenge, with its transmission influenced by climate variability, rapid urbanization, and the effectiveness of control interventions. In this study, we develop and analyze a mathematical model of dengue transmission incorporating parameter imprecision to account for uncertainties arising from environmental variability as key mosquito entomological parameters are

temperature-dependent and fluctuations in temperature introduce uncertainty into mosquito recruitment, mortality, and disease transmission rates. The existence and local stability of the disease-free and endemic equilibria are established, and the basic reproduction number is derived to characterize disease persistence. A normalized forward sensitivity analysis identifies the key parameters influencing transmission. Furthermore, insecticide application and treatment are incorporated as control measures within an optimal control framework. An objective functional is formulated to minimize infected human and mosquito populations while accounting for intervention costs. The cost-effectiveness of the control strategies is evaluated using the infected averted ratio (IAR), average cost-effectiveness ratio (ACER), and incremental cost-effectiveness ratio (ICER). Finally, MATLAB-based numerical simulations illustrate the analytical results and evaluate the effectiveness and economic efficiency of the proposed interventions. The findings demonstrate the importance of incorporating parameter imprecision when designing effective and economically sustainable dengue control strategies.

73. Weyl-type operators on weighted Bergman spaces: Berezin range and Berezin number

[C-73]

Somdatta Barik

Jadavpur University, India

Abstract: The study of isometric and unitary weighted composition operators on reproducing kernel Hilbert spaces of holomorphic functions has received considerable attention over the years. In this talk, we introduce a new class of weighted composition operators on weighted Bergman spaces, called Weyl-type weighted composition operators. We investigate their fundamental properties and determine their Berezin range and Berezin number. Finally, we obtain a class of operators that satisfy the reverse power inequality for the Berezin number.

74. On an Improvement of the Blanco-Koldobsky-Turnšek Characterization of Isometries via Level Vectors

[C-74]

Jayanta Manna

Jadavpur University, India

We improve the Blanco-Koldobsky-Turnšek characterization of isometries in normed linear spaces using the concept of level vectors of an operator. We characterize level vectors in terms of the preservation of Birkhoff-James orthogonality in a given direction and study the geometric and structural properties arising from them. Moreover, for spaces whose unit balls

have the Krein-Milman property, we obtain a further refinement of the Blanco-Koldobsky-Turnšek characterization of isometries.

75. Insight of Vector's Feeding Affinity in Zoonotic Cutaneous Leishmaniasis Transmission: A Case Study [C-75]

Homagnic Bose

Department of Mathematics, Jadavpur University, India

Abstract: Zoonotic Cutaneous Leishmaniasis (ZCL) is a complex vector-borne disease caused by Leishmania parasites which affects millions of people worldwide, making it a significant public health challenge nowadays. Investigating the role of animal reservoirs in disease propagation and the feeding affinity of vectors towards infected hosts remain quite challenging for the researchers. To address these complexities, our research develops an eight-dimensional deterministic human-vector-reservoir mathematical model that incorporates vector's feeding affinity for infected populations, offering a more realistic and detailed framework for understanding ZCL transmission. The feasibility of endemic equilibria is discussed, revealing a critical threshold value for vector's feeding affinity (ω^*V). Conditions for the persistence or eradication of the disease emphasizing the importance of (ω^*V) is evaluated through bifurcation analysis. Moreover, to validate the model, incidence data from a ZCL-endemic region in the middle east are utilized. Model parameters are estimated using Markov Chain Monte Carlo techniques. Key findings include a threshold feeding affinity range of (4.75, 8.25) and Maximum A Posteriori (MAP) estimate of 5.85. Our findings demonstrate the significant role of vector's feeding affinity in ZCL transmission and a case study to highlight the practical relevance of the theoretical findings.

76. An Adaptive Hybrid Framework for Spatiotemporal Forecasting and Risk Assessment of Cutaneous Leishmaniasis in Iran [C-76]

Satyajit Mukherjee

Department of Mathematics, Jadavpur University, India

Abstract: Cutaneous leishmaniasis (CL) remains an important public-health challenge in Iran, where disease incidence exhibits substantial spatial and temporal heterogeneity. We develop a modular adaptive hybrid framework for forecasting CL incidence while explicitly separating temporal persistence, environmental effects, and spatial structure. Using province-level data from 31 provinces during 2013–2020, the framework integrates chronological rolling-origin validation, statistical and machine-learning environmental models, temporal forecasting, spatial risk modelling, and bootstrap-based uncertainty and stability assessment. The final two years (2019–2020) are retained as a protected out-of-sample test set and are not used for model selection or adaptive weighting. Temporal persistence provides the strongest and most reproducible forecasting signal, with an autoregressive AR(1) model selected as the temporal backbone. Although CatBoost captures a modest environmental signal during development,

bootstrap analysis indicates insufficient stability for environmental admission into the primary forecast. Accordingly, the adaptive integration assigns zero weight to the environmental branch rather than forcing a hybrid contribution. Spatial modelling provides an independent relative-risk structure, complementing the temporal forecast. Our architecture therefore demonstrates an evidence-driven approach in which model complexity is introduced only when it provides reproducible incremental predictive value, offering a transparent framework for epidemiological forecasting and spatial risk assessment of CL.

77. A Graph-Based Mathematical Framework for Modelling Neurodegenerative Disease States from Patient-Level Data

[C-77]

Anirban Tarafdar

Abstract: Neurodegenerative disorders exhibit substantial heterogeneity across individuals, making binary disease labels insufficient to describe the underlying continuum of disease-associated states. This study proposes a graph-based mathematical framework for modelling neurodegenerative disease states from patient-level data. Each patient is represented as a node, while pairwise similarities between patients define weighted interactions and form a patient-similarity network. A continuous disease-state function is then formulated on this network through an energy-based mathematical model that combines structural smoothness with consistency with observed disease conditions. The resulting formulation assigns each patient a continuous state ranging from healthy-associated through intermediate to disease-associated regions of the modelled state space. The mathematical properties and stability of the estimated disease-state function are investigated using network-based and statistical analyses. The framework will be evaluated on independent datasets representing Parkinson's disease and Alzheimer's disease to examine whether the same mathematical formulation can characterize disease-associated structure across different neurodegenerative disorders. Where clinical severity measures are available, the relationship between the estimated disease state and disease severity will also be examined. Machine-learning classification may be used as a secondary validation to determine whether the proposed mathematical representation provides information beyond conventional patient-level features. The study aims to develop an interpretable mathematical modelling framework for analysing heterogeneous disease states in complex neurodegenerative systems.

78. An Overview of Complex Kleinian Groups

[C-78]

Rahul Mandal

IISER Mohali, India

Abstract: Classical Kleinian groups are discrete subgroups of $PSL(2, \mathbb{C})$ acting projectively on the Riemann sphere $\hat{\mathbb{C}}$. When generalizing to higher complex dimensions via $PSL(n+1, \mathbb{C})$, the classical notion of pointwise orbit accumulation fails. In this talk, we present an overview of complex Kleinian groups,

focusing on R. S. Kulkarni's (1978) foundational framework for defining the limit set $\Lambda_K(\Gamma)$ and domain of discontinuity $\Omega_K(\Gamma)$, and illustrating how to explicitly compute these limit sets for cyclic subgroups in $\mathrm{PSL}(3, \mathbb{C}) \curvearrowright \mathbb{P}^2_{\mathbb{C}}$.

Poster Presentations

1. Impossibility in Constructing 1 Radian Explained - An Approach to Better Approximation [PP-1]

Narendranath Mitra

Department of Mathematics, Jadavpur University, Kolkata, India

Abstract: Lindemann's theorem about the transcendence of numbers and in the context of Field theory result that every constructible number is algebraic over $\mathbb{Q}$ suggest that 1 radian is inconstructible by ruler and compass. Therefore, our construction of 1 radian with ruler and compass precisely with a minimal deviation of 0.00000009 ($9/10^8$) radian is mentionworthy.

2. Model Discovery in Structurally Sensitive Nonlinear Systems Using a Data-Driven Approach [PP-2]

Subham Kandyal

Abstract: Structural sensitivity refers to the phenomenon in which nearly indistinguishable functional forms can produce significantly different dynamical behavior. This creates an important challenge in biological modeling because the true functional form governing species interactions is generally unknown in resource-consumer systems. Previous studies have largely examined structural sensitivity by comparing prescribed functional forms and their resulting stability. Here, we explore whether the governing interaction structure can instead be identified directly from observations prior to the system undergoing a bifurcation. We propose SIFT-iKAN (System Identification and Functional-response Testing with an integral Kolmogorov-Arnold Network), a data-driven framework that simultaneously identifies the functional response and recovers the underlying dynamical equations from time-series data. We test the framework on synthetic trajectories generated from the Rosenzweig-MacArthur (R-M) model, taking each of five nearly indistinguishable functional responses as the true response in separate experiments. We further test the framework across different noise levels to assess its robustness. The results show that our proposed framework can successfully identify the governing functional response and recover the governing equations even from noisy data, providing a data-driven way to detect structural sensitivity.

3. Impact of Nanoparticles Dispersion on Entropy Generation in a Microchannel Poiseuille Flows: A Mathematical Model [PP-3]

Ramotjaki Lucky Monaedi

Stellenbosch University, South Africa

Abstract: Recent advent of nanotechnology has greatly improved the ability to augment heat transfer processes through innovative mixture of conventional fluid with nano-sized solid metallic or non-metallic particles (1–100nm diameters) known as nanoparticles. In this paper the impact of Copper nanoparticles dispersion on inherent irreversibility in a microchannel Poiseuille flow of a water base nanofluid is examined. The governing nonlinear differential equations that rely on modified Buongiorno's nanofluid model is proposed and numerically tackled via shooting method coupled with Runge-Kutta-Fehlberg integration scheme. Pertinent results showing the effects of emerging thermophysical parameters on the nanofluid velocity, temperature, nanoparticles concentration, skin friction, Nusselt number, Sherwood number, entropy generation rate and Bejan number are presented graphically and discussed. It is found that thermophoresis of mobile nanoparticles escalates entropy production rate while nanoparticles injection and the Brownian motion lessen it.

4. Biomathematics in Cancer Research: Looking into how Mathematical Models Are Used to Understand Tumor Growth and the Effectiveness of Different Treatment Strategies [PP-4]

Subham Mondal

Jadavpur University

Abstract: This poster explores the critical role of biomathematics in cancer research, focusing on how mathematical models enhance our understanding of tumor growth and optimize treatment strategies. It begins with an overview of cancer biology, highlighting the complexities of tumor dynamics and the molecular mechanisms driving growth. It then delves into various mathematical modeling approaches, including Deterministic illustrating their applications in predicting tumor behavior and treatment responses. Case studies demonstrate the real-world impact of these models on optimizing chemotherapy, radiation therapy, and immunotherapy. Challenges such as data availability, model complexity, and clinical translation are discussed, alongside future directions in the field, including advances in computational power and personalized medicine. Ultimately, this emphasizes the transformative potential of biomathematics in improving cancer treatment outcomes and underscores the importance of interdisciplinary collaboration in advancing the field.

